Integral Hodge Conjecture – Invariance Under Twisting

Contents

Abstract

We establish a structural classification of the role of gerbe twists in the integral Hodge conjecture. First, we prove that twisting generalized cohomology theories by $[H] \in H^3(X;\mathbb{Z})$ produces no new topological obstructions to the untwisted conjecture: any multiplicative cycle-receiving theory is necessarily complex-oriented and factors through MU, while twisted K-theory fails this criterion because $d_3^{[H]}(1) = [H] \neq 0$. Furthermore, by Hertl's ghost theorem, every H^3-twist of MU leaves the primary differential $d_3^{MU,\tau} = \mathrm{Sq}^3_{\mathbb{Z}}$ invariant. Any remaining obstruction to the classical conjecture is shown to be purely algebro-geometric, residing in the cokernel of the Levine-Morel realization $\Omega^i(X) \to MU^{2i}(X(\mathbb{C})) \otimes_{MU^*} \mathbb{Z}$. Finally, we transfer the framework of this paper to where the twisted differential *does* govern: the integral Hodge conjecture for α-twisted sheaves, where the missing unit is quantitatively replaced by the index $\operatorname{ind}(\alpha)$.

Let X be a smooth projective variety. The cycle map

$$\mathrm{cl}_i : \mathrm{CH}^i(X) \to H^{2i}(X;\mathbb{Z})$$

sends (the class of)[1] a codimension-i subvariety $Z \subseteq X$ to the Poincaré dual of its fundamental class $[Z] \in H_{2n-2i}(X;\mathbb{Z})$, where $n = \dim X$. The integral Hodge conjecture (in codimension i) states that the cycle map is surjective on

$$\mathrm{Hdg}^{2i}(X;\mathbb{Z}) = \left\{\alpha \in H^{2i}(X;\mathbb{Z}) \;\middle|\; \alpha_{\mathbb{C}} \in H^{i,i}(X)\right\} = H^{2i}(X;\mathbb{Z}) \cap H^{i,i}(X)$$

that is, $\operatorname{im}(\mathrm{cl}_i) = \mathrm{Hdg}^{2i}(X;\mathbb{Z})$. The impetus for this conjecture stems from the $i = 1$ case, where by the Lefschetz theorem on $(1,1)$-classes $\operatorname{im}\mathrm{cl}_1 = \mathrm{Hdg}^2(X;\mathbb{Z})$ [18, Ch. 1]. Atiyah and Hirzebruch in [15] showed that it fails for $i = 2$. They factor the cycle map through K-theory,

$$\mathrm{CH}^i(X) \to KU^0(X) \xrightarrow{\theta} H^{2i}(X;\mathbb{Z})$$

with θ an edge homomorphism of the Atiyah-Hirzebruch spectral sequence (section 1 recalls the construction, including the filtration on which θ is defined), so a Hodge class outside the image of θ cannot be algebraic. Such classes do exist. Every torsion class is a Hodge class, vanishing in $H^{2i}(X;\mathbb{C})$, and the image of θ is cut out by cohomology operations: Atiyah and Hirzebruch produce smooth projective varieties, Godeaux-Serre quotients for finite groups, carrying a torsion class of degree four on which the degree-three operation $\mathrm{Sq}^3_{\mathbb{Z}}$ recalled in section 1 is nonzero. The resulting *K-theory obstruction map* is the composite

$$o^i_{KU} : \mathrm{Hdg}^{2i}(X;\mathbb{Z}) \hookrightarrow H^{2i}(X;\mathbb{Z}) \twoheadrightarrow \operatorname{coker}\theta$$

and $o^i_{KU}(\beta) \neq 0$ forces β to be non-algebraic.

Two questions remain on the integral side: for which smooth projective varieties X does the integral Hodge conjecture hold, and how finely can the failure be measured. Totaro in [2] addressed both, by defining smooth projective approximations of the topological spaces BG and by refining the factorization through cobordism:

$$\mathrm{CH}^i(X) \xrightarrow{\widetilde{\mathrm{cl}}} MU^{2i}(X) \otimes_{MU^*} \mathbb{Z} \xrightarrow{\theta} H^{2i}(X;\mathbb{Z}), \tag{1}$$

where θ is the edge homomorphism of the Atiyah-Hirzebruch spectral sequence and $\widetilde{\mathrm{cl}}$ is defined on a subvariety Z by resolving it, $\xi\colon \widetilde{Z} \to X$, and taking $\xi_*(1)$. Totaro showed that a Hodge class that fails to lift along θ is not algebraic. The mechanism behind both refinements is the same: KU carries a complex orientation, $c_1^{KU}(L) = [L] - 1$, equivalently a ring map $MU \to KU$, and it is the orientation that produces the classes $\xi_*(1)$. Twisted K-theory carries no such orientation (section 3) and therefore appears to lie outside the range of (1), which is what makes it a candidate for a finer obstruction. We show that it is not one. A twist can enter (1) at three points, and each case is closed in this paper:

1. *No twisted theory receives the cycle.* Any homotopy-commutative ring spectrum carrying an intersection-compatible cycle class is complex-oriented, and its cycle classes are the Gysin classes of that orientation (theorem 3.2). Thus, no such theory refines cl_i beyond MU. For $[H] \neq 0$ the group $KU^0_{[H]}(X)$ carries no class that can play the role of the fundamental class of X, and twisted K-theory carries no such structure (proposition 3.7).

[1] the assignment descends from the group $Z^i(X)$ of codimension-i cycles to $\mathrm{CH}^i(X)$ since rationally equivalent cycles are homologous

2. *Twisting the universal theory does not move its differential.* Twisting MU produces a spectrum MU_τ (definition 4.1), but by a theorem of Hertl [22] every H^3-twist of MU is a *ghost*, and the primary differential of the twisted spectral sequence is the untwisted $\mathrm{Sq}^3_{\mathbb{Z}}$ (theorem 4.3).

3. *The bare twisted differential decides nothing new.* Read as an operation on integral cohomology, $d_3^{[H]} = \mathrm{Sq}^3_{\mathbb{Z}} + [H] \smile (-)$ annihilates the algebraic multiples of a class α exactly when a divisibility $N \big| d$ holds, where d records which multiples of α are algebraic (proposition 4.10). On $B(\mathrm{SL}_6/\mu_2)$ this makes the condition on the codimension-three Chern class c_3 equivalent to the non-algebraicity of c_3 (proposition 5.8).

Section 6 then focuses on the map $\widetilde{\mathrm{cl}}$ and the cokernel of the realization of Levine-Morel algebraic cobordism into MU, an invariant of the algebraic side. Section 7 records the problem for which the twisted differential *is* the governing obstruction, the integral Hodge conjecture for α-twisted sheaves. There, the objects tested are themselves twisted, the missing unit is replaced by the index of the Brauer class, and the resulting conjecture is logically independent of the one explored in this paper.

The three factorizations studied in this paper, and their primary differentials, are collected bellow, with $\theta_{[H]}$ the edge homomorphism of the twisted spectral sequence of section 2:

untwisted (section 1):	$\mathrm{CH}^i(X) \xrightarrow{\widetilde{\mathrm{cl}}} MU^{2i}(X) \otimes_{MU^*} \mathbb{Z} \xrightarrow{\theta} H^{2i}(X;\mathbb{Z}),$	$d_3 = \mathrm{Sq}^3_{\mathbb{Z}}$
twisted attempt (section 3):	$\mathrm{CH}^i(X) \dashrightarrow KU^0_{[H]}(X) \xrightarrow{\theta_{[H]}} H^{2i}(X;\mathbb{Z}),$	$d_3^{[H]} = \mathrm{Sq}^3_{\mathbb{Z}} + [H] \smile (-)$
twisted sheaves (section 7):	$K_0(X,\alpha) \longrightarrow KU^0_{[H]}(X) \xrightarrow{\theta_{[H]}} H^{2i}(X;\mathbb{Z}),$	$d_3^{[H]}$ affects

The dashed arrow does not exist (proposition 3.7), and the mechanism of its failure is the substantive work of this paper.

Conventions

For the Hodge-theoretic statements, X is either a smooth projective complex variety or one of Totaro's smooth projective approximations of a classifying space BG [2]. These are finite-dimensional, so the Atiyah-Hirzebruch spectral sequences below converge strongly. The structural results of sections 3, 4 and 7 quantify instead over all smooth quasi-projective varieties over $\mathbb{C}$, as their proofs use total spaces of vector bundles, which are quasi-projective and not projective. Subvarieties are taken to be irreducible, so smooth closed subvarieties are connected, and cycle-level constructions extend additively.

On an approximation of BG every class of H^{2i} is a Hodge class: the torsion is automatically Hodge, vanishing in $H^{2i}(BG;\mathbb{C})$, and the free part is of type (i,i) because the cohomology of the classifying space of a complex linear algebraic group is of Tate type [2]. Hence $\mathrm{Hdg}^{2i}(BG;\mathbb{Z}) = H^{2i}(BG;\mathbb{Z})$ and the integral Hodge conjecture there is the surjectivity of $\mathrm{cl}_i\colon \mathrm{CH}^i(BG) \to H^{2i}(BG;\mathbb{Z})$. Both cl_i and the spectral-sequence differentials acting on $H^{2i}(BG;\mathbb{Z})$ are defined by stability on the approximations, taking their common value on X_m once m exceeds the cohomological degrees involved (for the d_3-statements in degree $2i$, any $m > 2i+3$ serves). This is the sense in which the example of section 5, though phrased on the infinite-dimensional BG, is to be read.

1 The Untwisted Obstruction and Its Factorization

Take the cycle map $\mathrm{cl}_i\colon \mathrm{CH}^i(X) \to H^{2i}(X;\mathbb{Z})$ of the introduction (see [35] for the construction). To locate the failure of surjectivity onto Hdg^{2i} one factors it through a third group whose own lifting problem is computable: Atiyah and Hirzebruch used KU^0, and Totaro refined it to $MU^{2i} \otimes_{MU^*} \mathbb{Z}$,

$$\mathrm{CH}^i(X) \to KU^0(X) \xrightarrow{\theta} H^{2i}(X;\mathbb{Z}), \qquad \mathrm{CH}^i(X) \to MU^{2i}(X) \otimes_{MU^*} \mathbb{Z} \xrightarrow{\theta} H^{2i}(X;\mathbb{Z})$$

In both, the right-hand map is an edge homomorphism of an Atiyah-Hirzebruch spectral sequence, and a Hodge class is obstructed when it fails to pass the differentials of that sequence. This section recalls the spectral sequence for KU and the identification of its first differential with $\mathrm{Sq}^3_{\mathbb{Z}}$. Section 2 twists them.

For a generalized cohomology theory E^* and X finite-dimensional (the conventions above, which is what makes the sequence converge strongly), the Atiyah-Hirzebruch spectral sequence reads

$$E_2^{p,q} = H^p(X;\, E^q(\mathrm{pt})) \Longrightarrow E^{p+q}(X), \qquad d_r : E_r^{p,q} \to E_r^{p+r,\, q-r+1}$$

We compute with $E = ku$, the connective cover of KU, since for finite-dimensional X as in the Conventions the map $ku^0(X) \to KU^0(X)$ is an isomorphism, since the cofibre of $ku \to KU$ has homotopy concentrated in degrees ≤ -2, whose contributions to total degrees 0 and -1 would come from columns $p < 0$. By Bott periodicity $KU^q(\mathrm{pt}) = \mathbb{Z}$ for q even and 0 for q odd, and connectivity truncates the positive rows[2], so

$$ku^q(\mathrm{pt}) = \begin{cases} \mathbb{Z} & q \text{ even and } q \leq 0 \\ 0 & \text{otherwise} \end{cases}$$

and the E_2-page is a copy of the integral cohomology of X in each even row $q \leq 0$ and zero elsewhere:

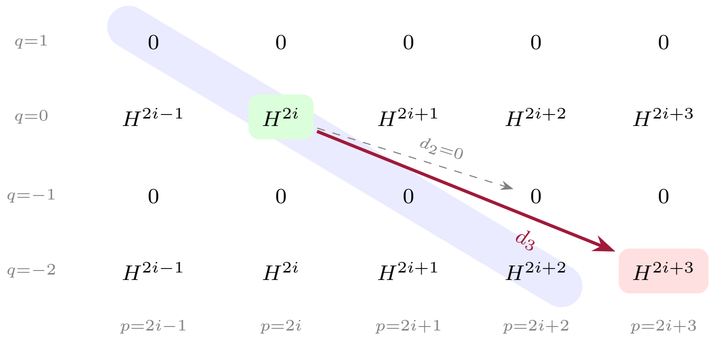

Figure 1: The page $E_2 = E_3$. A Hodge class lives in the green entry $E_2^{2i,0} = H^{2i}(X;\mathbb{Z})$. Its first obstruction d_3 lands in the red entry $H^{2i+3}(X;\mathbb{Z})$. Even differentials (dashed) die in the odd, zero rows. The shaded diagonal $p+q = 2i$ collects the graded pieces of $ku^{2i}(X)$, whose image in $KU^{2i}(X) \cong KU^0(X)$ is the filtration piece $F^{2i}KU^0(X)$ on which the edge map is defined.

The differential d_r lowers q by $r-1$, so every even d_r lands in an odd row and vanishes, in particular $d_2 = 0$ and $E_3 = E_2$. The first differential that can act on a class $\beta \in H^{2i}(X;\mathbb{Z})$, which sits in the corner $E_2^{2i,0}$, is therefore

$$d_3 : E_3^{2i,0} = H^{2i}(X;\mathbb{Z}) \to H^{2i+3}(X;\mathbb{Z})$$

the remaining ones being $d_5, d_7, \cdots$. The class β lifts to $KU^0(X)$ only if all of them vanish on it.

A differential d_r landing at $(2i, 0)$ starts at $(p,q) = (2i-r,\, r-1)$, and $r \geq 3$ puts $q = r-1 \geq 2 > 0$, a row where the connective ku vanishes. Hence $E_\infty^{2i,0} = \bigcap_r \ker d_r$ is a subgroup of $E_2^{2i,0} = H^{2i}(X;\mathbb{Z})$. The same vanishing rows put every graded piece of $ku^{2i}(X)$ in filtration $\geq 2i$, so the composite

$$ku^{2i}(X) = F^{2i}ku^{2i}(X) \twoheadrightarrow F^{2i}ku^{2i}(X)/F^{2i+1} = E_\infty^{2i,0} \hookrightarrow E_2^{2i,0} = H^{2i}(X;\mathbb{Z})$$

[2]in the cohomological grading $ku^q(\mathrm{pt}) = \pi_{-q}ku$

is the edge homomorphism

$$\theta\colon ku^{2i}(X) \to H^{2i}(X;\mathbb{Z})$$

defined on the whole connective group. On $KU^0(X) = ku^0(X)$ it is only partially defined: under $ku^{2i}(X) \to KU^{2i}(X) \cong KU^0(X)$ the connective group maps onto the skeletal filtration subgroup $F^{2i}KU^0(X)$ of classes vanishing on the $(2i-1)$-skeleton (a map from a $(2i-1)$-complex to the $(2i-1)$-connected space $BU\langle 2i\rangle$ is null-homotopic, and the converse lifting is standard obstruction theory), so a class of $F^{2i}KU^0(X)$ enters the domain of θ through a choice of connective lift. The choice is harmless, as the obstruction statements below depend only on the image of θ, which consists of permanent cycles. Rationally the sequence degenerates, meaning that every differential vanishes after $\otimes\mathbb{Q}$ and $E_2 \otimes \mathbb{Q} = E_\infty \otimes \mathbb{Q}$ (rationally ku splits as a wedge of Eilenberg-MacLane spectra, which have no k-invariants). Under this degeneration $\theta \otimes \mathbb{Q}$ is the restriction to F^{2i} of the degree-$2i$ component ch_i of the Chern character $\mathrm{ch}\colon KU^0(X)\otimes\mathbb{Q} \xrightarrow{\sim} \bigoplus_i H^{2i}(X;\mathbb{Q})$ [27], whose values on the sheaf classes $[\mathcal{O}_Z]$ below involve the exponential series and so carry denominators. Passing to θ is what retains the torsion and the subgroup index that the rational isomorphism discards.

The algebraic cycle enters through this filtration. A codimension-i subvariety $Z \subseteq X$ gives the coherent sheaf $\mathcal{O}_Z$, which on the smooth X has a finite resolution by vector bundles, hence a class $[\mathcal{O}_Z] \in KU^0(X)$. Extending additively gives a homomorphism $Z^i(X) \to KU^0(X)$ on the group of cycles. The resolution is exact off Z, so the class restricts to zero on $X \setminus Z$, hence on the $(2i-1)$-skeleton of a cell structure dual to Z (such a skeleton lies in $X \setminus Z$, as Z has real codimension $2i$). Therefore $[\mathcal{O}_Z] \in F^{2i}KU^0(X)$, and so is in the domain of θ, and $\theta([\mathcal{O}_Z]) = \mathrm{cl}_i(Z)$ [15] (for smooth Z this is the Thom-class comparison of section 3). The composite $Z^i(X) \to F^{2i}KU^0(X) \xrightarrow{\theta} H^{2i}(X;\mathbb{Z})$ is therefore the cycle map on cycles, so it kills rational equivalence and yields

$$\mathrm{im}(\mathrm{cl}_i) \ \subseteq\ \mathrm{im}(\theta) \ \subseteq\ H^{2i}(X;\mathbb{Z})$$

which is the factorization used throughout. Note from this that a Hodge class outside $\mathrm{im}\theta$ is not algebraic. On $\mathrm{CH}^i(X)$ itself the KU^0-valued map is defined only up to classes supported in codimension $> i$, and the obstruction statement uses images alone.

The first obstruction to lifting along θ is d_3, which Atiyah and Hirzebruch identify with the *integral Steenrod operation* $\mathrm{Sq}^3_{\mathbb{Z}}$. As d_3 is the map the twist deforms, its definition is recorded here. The $\mathbb{Z}/2\mathbb{Z}$-cohomology carries the *Steenrod squares*

$$\mathrm{Sq}^i : H^n(X;\mathbb{Z}/2\mathbb{Z}) \to H^{n+i}(X;\mathbb{Z}/2\mathbb{Z})$$

natural in X and stable, characterised by $\mathrm{Sq}^0 = \mathrm{id}$, $\mathrm{Sq}^i(\alpha) = 0$ for $i > |\alpha|$, the top square $\mathrm{Sq}^{|\alpha|}(\alpha) = \alpha \smile \alpha \in H^{2|\alpha|}(X;\mathbb{Z}/2\mathbb{Z})$[3], together with the Cartan and Adem relations [21]. The coefficient sequence $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ gives for any CW-complex X the long exact sequence

$$\cdots \to H^n(X;\mathbb{Z}) \xrightarrow{\times 2} H^n(X;\mathbb{Z}) \xrightarrow{\rho_2} H^n(X;\mathbb{Z}/2\mathbb{Z}) \xrightarrow{\delta} H^{n+1}(X;\mathbb{Z}) \to \cdots$$

with ρ_2 reduction mod 2 and δ the integral Bockstein. The integral Steenrod operation is the composite

$$\mathrm{Sq}^3_{\mathbb{Z}} = \delta \circ \mathrm{Sq}^2 \circ \rho_2 : H^n(X;\mathbb{Z}) \to H^{n+3}(X;\mathbb{Z})$$

that is, reduce mod 2, apply Sq^2, and lift back to $\mathbb{Z}$ by the Bockstein. It is a stable operation of degree 3, and its image is 2-torsion, lying in the image of δ. Atiyah and Hirzebruch prove $d_3 = \mathrm{Sq}^3_{\mathbb{Z}}$ [15]: the differential d_3 is a stable operation $H^n(-;\mathbb{Z}) \to H^{n+3}(-;\mathbb{Z})$, the first k-invariant of ku, the group of stable degree-three integral operations is $\mathbb{Z}/2\mathbb{Z}$, generated by $\mathrm{Sq}^3_{\mathbb{Z}}$, and the zero operation is ruled out by exhibiting spaces on which d_3 acts nontrivially.

[3] the squaring property holds only in top degree, and throughout with $\mathbb{Z}/2\mathbb{Z}$-coefficients

2 The Twist and Its Differential

A rank-n vector bundle $E \to X$ has a projectivization $\mathbb{P}(E)$, a bundle of projective spaces with structure group $\mathrm{PGL}_n(\mathbb{C})$. Not every PGL_n-bundle arises this way, and the twisted bundles are the objects added to make it so. The failure is governed by the central extension

$$1 \longrightarrow \mathbb{C}^* \longrightarrow \mathrm{GL}_n(\mathbb{C}) \longrightarrow \mathrm{PGL}_n(\mathbb{C}) \longrightarrow 1$$

whose centrality lets the nonabelian cohomology sequence continue one step past H^1:

$$H^1(X; \underline{\mathrm{GL}_n(\mathbb{C})}) \xrightarrow{q} H^1(X; \underline{\mathrm{PGL}_n(\mathbb{C})}) \xrightarrow{\partial} H^2(X; \underline{\mathbb{C}^*})$$

where $\underline{\mathbb{C}^*}$ is the sheaf of nowhere-vanishing $\mathbb{C}$-valued functions (continuous in the analytic setting, regular in the étale one). A PGL_n-bundle P is the projectivization of a GL_n-bundle exactly when $\partial(P) = 0$. The exponential sequence $0 \to \mathbb{Z} \to \underline{\mathbb{C}} \xrightarrow{\exp} \underline{\mathbb{C}^*} \to 0$ records this obstruction in ordinary cohomology through its connecting homomorphism, the exponential Bockstein $\delta\colon H^2(X; \underline{\mathbb{C}^*}) \to H^3(X; \mathbb{Z})$, and

$$[H] \;:=\; \delta\big(\partial(P)\big) \;\in\; H^3(X; \mathbb{Z})$$

is the *gerbe class* of P.

Finite rank confines the twists to the torsion subgroup $H^3(X; \mathbb{Z})_{\mathrm{tors}}$. Recall that one cohomology degree lower, $H^2(X; \mathbb{Z})$ is the classification of line bundles: the connecting map $H^1(X; \underline{\mathbb{C}^*}) \to H^2(X; \mathbb{Z})$ is c_1 which is an isomorphism for the sheaf of continuous functions. As PGL_n is also the quotient of SL_n by its centre μ_n, the obstruction of a PGL_n-bundle is the image of a class in $H^2(X; \mu_n)$, hence killed by n.

To see the objects the twist adds, pass to Čech cocycles on a cover $\{U_i\}$ trivializing the data. A rank-n vector bundle is transition data $g_{ij} \in \mathrm{GL}_n(\mathcal{O}_X(U_{ij}))$ satisfying the cocycle condition

$$g_{ij} g_{jk} g_{ki} = \mathrm{id}$$

on triple intersections. Let $h = (h_{ijk})$ be a Čech 2-cocycle valued in $\mathcal{O}_X^*$ representing $\partial(P)$, so that $[H] = \delta[h]$. An $[H]$-*twisted vector bundle* of rank n is transition data whose cocycle condition holds only up to h:

$$g_{ij} g_{jk} g_{ki} = h_{ijk} \cdot \mathrm{id} \;\in\; Z(\mathrm{GL}_n\big(\mathcal{O}_X(U_i \cap U_j \cap U_k)\big))$$

the untwisted bundles being the $h = 1$ case. Since h_{ijk} is central, the projectivized data $\overline{g}_{ij} \in \mathrm{PGL}_n$ satisfies the strict cocycle condition, so an $[H]$-twisted rank-n bundle is a linearization of a PGL_n-bundle with gerbe class $[H]$. Every PGL_n-bundle is the projectivization of a twisted bundle, whereas only those with $[H] = 0$ were projectivizations of untwisted ones. This is the sense in which twisting enlarges the supply of geometric objects.

The same objects have an algebra-theoretic description, used in section 7.

Definition 2.1: Azumaya Algebra and Brauer Class

An *Azumaya algebra of degree* n on X is a sheaf $\mathcal{A}$ of $\mathcal{O}_X$-algebras, locally free of rank n^2 as an $\mathcal{O}_X$-module, whose fibres are matrix algebras M_n. Equivalently, $\mathcal{A}$ is locally isomorphic to $M_n(\mathcal{O}_X)$ in the topology used. Two Azumaya algebras are *Brauer equivalent* if $\mathcal{A} \otimes \mathcal{E}nd(E) \cong \mathcal{A}' \otimes \mathcal{E}nd(E')$ for vector bundles E, E', and the equivalence classes form the *Brauer group* $\mathrm{Br}(X)$ under $\otimes$. The class $\alpha = [\mathcal{A}] \in \mathrm{Br}(X)$ has a gerbe class $[H] \in H^3(X; \mathbb{Z})_{\mathrm{tors}}$, obtained as above from the PGL_n-bundle of local trivializations of $\mathcal{A}$.

For an $[H]$-twisted bundle E the factors h_{ijk} cancel in $\mathcal{E}nd(E)$ (as $\mathcal{E}nd(E) \cong E \otimes E^*$), which is therefore an (untwisted) sheaf of algebras, Azumaya of degree rk$E$ and of Brauer class $[H]$. Conversely, the $[H]$-twisted bundles are exactly the modules over an Azumaya algebra of that class ([12], and [34, §2.2.2] for the equivalence in general).

Direct sum preserves the twist, so the $[H]$-twisted bundles form an additive category $\mathrm{Vect}_{[H]}(X)$ with $[E] + [E'] = [E \oplus E']$, and

$$K^0_{[H]}(X) = K_0\big(\mathrm{Vect}_{[H]}(X)\big)$$

is *twisted K-theory.* Tensor product does not preserve the twist: the transition data of $E \otimes E'$ has cocycle $h_{ijk}h'_{ijk}$, so

$$E \in \mathrm{Vect}_{[H]}(X), \quad E' \in \mathrm{Vect}_{[H']}(X) \quad \Longrightarrow \quad E \otimes E' \in \mathrm{Vect}_{[H]+[H']}(X)$$

Hence $K^0_{[H]}(X)$ is not a ring. What the tensor product gives is a system of pairings

$$K^0_{[H]}(X) \otimes K^0_{[H']}(X) \longrightarrow K^0_{[H]+[H']}(X), \tag{2}$$

associative and commutative, which make the direct sum $K^0_\bullet(X) = \bigoplus_{\eta \in H^3(X;\mathbb{Z})} K^0_\eta(X)$ an $H^3(X;\mathbb{Z})$-graded commutative ring with identity the class of the trivial line bundle in degree 0. Only the torsion η admit twisted bundles of finite rank, as shown above, so only they carry nonzero summands, and for torsion twists this bundle-theoretic model agrees with the operator-theoretic twisted K-theory of Atiyah-Segal [7]. The special case $[H'] = 0$ of (2) makes each $K^0_{[H]}(X)$ a $K^0(X)$-module.

The grading is realized topologically by "projective Hilbert bundles": the group $H^3(X;\mathbb{Z})$ classifies twists, and the tensor product of bundles realizes the addition of classes as follows. A *projective Hilbert bundle* is a fibre bundle with fibre $\mathbb{P}(\mathcal{H})$, for $\mathcal{H}$ a separable infinite-dimensional Hilbert space, and structure group $\mathrm{PU}(\mathcal{H}) = U(\mathcal{H})/U(1)$. By Kuiper's theorem $U(\mathcal{H})$ is contractible [28], so $B\mathrm{PU}(\mathcal{H}) \simeq B^2U(1) \simeq K(\mathbb{Z}, 3)$, and isomorphism classes of projective Hilbert bundles on X are in bijection with $H^3(X;\mathbb{Z})$. A choice of unitary isomorphism $\mathcal{H} \otimes \mathcal{H} \cong \mathcal{H}$ induces a product $\mathrm{PU}(\mathcal{H}) \times \mathrm{PU}(\mathcal{H}) \to \mathrm{PU}(\mathcal{H})$ modelling the H-space addition on $K(\mathbb{Z}, 3)$, so the tensor product of projective Hilbert bundles adds the corresponding classes [7, Prop. 2.1(ii), (iii), (vii)]. The pairing (2) is the finite-rank shadow of this addition.

Twisted K-theory is computed via the Atiyah-Segal spectral sequence that deforms the untwisted case. The twist is pulled back along the map $X \to K(\mathbb{Z}, 3)$ classifying $[H]$, and $K(\mathbb{Z}, 3)$ is 2-connected, so the local coefficient system is trivial: the page $E_2 = E_3$ agrees with the untwisted one, and only the differentials can move. The first differential that can act on a class of the corner $E_3^{2i,0}$ becomes

$$d_3^{[H]}(-) = \mathrm{Sq}^3_{\mathbb{Z}} + [H] \smile (-)$$

by a universal-operations argument [8, Prop. 4.6]. The higher differentials deform as well, by higher operations in the twist, a point section 4 returns to. To summarize the argument, the difference $d_3^{[H]} - \mathrm{Sq}^3_{\mathbb{Z}}$ is additive in its input x and natural in the pair (x, η), where $\eta \in H^3(-;\mathbb{Z})$ classifies the twist, hence is a class in

$$H^{p+3}(K(\mathbb{Z}, p) \times K(\mathbb{Z}, 3); \pi_2 KU)$$

the coefficients $\pi_2 KU \cong \mathbb{Z}$ recording the target row $q = -2$. As $H^4(K(\mathbb{Z}, 3);\mathbb{Z}) = H^5(K(\mathbb{Z}, 3);\mathbb{Z}) = 0$, the only term linear in x is $\lambda\,(\eta \smile x)$ with $\lambda \in \pi_2 KU = \mathbb{Z}$, and evaluating on $X = S^3$ where the twisted theory is the bundle of spectra clutched by the restriction of the twist to S^2, identifies λ with the generator. This computation is carried out in full, for an arbitrary ring spectrum in place of KU, in the proof of theorem 4.3, where the coefficient λ is exactly what separates KU from MU. So the twist deforms the first k-invariant of the KU-spectrum: where the untwisted attaching map contributes $\mathrm{Sq}^3_{\mathbb{Z}}$, the $[H]$-twisted spectrum adds cup product with the gerbe class. The sign of the cup term depends on the normalization of the twist (Atiyah and Segal write $\mathrm{Sq}^3_{\mathbb{Z}} - \eta \smile (-)$), and it affects neither the kernel nor the image, so we suppress it.

With twists established, the question now becomes:

Can a twist $[H]$ be chosen so that $d_3^{[H]}\beta = 0$ detects some Hodge classes?

We show that either it cannot, or the classes it detects are exactly those whose algebraicity was already the open question.

3 Complex Orientation and the Cycle-Receiving Theorem

The twisted differential $d_3^{[H]} = \mathrm{Sq}^3_{\mathbb{Z}} + [H] \smile (-)$ obstructs *algebraicity* only if it vanishes on algebraic classes. The natural way to prove that it does is to have the twisted theory *receive* the cycle, exactly as KU and MU do:

$$\mathrm{CH}^i(X) \longrightarrow KU^0_{[H]}(X) \longrightarrow H^{2i}(X;\mathbb{Z})$$

This section proves that no such factorization exists: through the work of Panin and Smirnov, a cycle class compatible with intersection products and the fundamental-class normalisation *is* an orientation, and twisted K-theory carries none. The weaker question of an additive lift is taken up in section 4.

Throughout, $E^*(X)$ abbreviates $E^*(X(\mathbb{C}))$ for X a smooth quasi-projective variety over $\mathbb{C}$, and $F^\bullet$ denotes the skeletal filtration on $E^*(X(\mathbb{C}))$ with associated Atiyah-Hirzebruch spectral sequence $E_2^{p,q} = H^p(X;E^q)$. The hypothesis of a homotopy-commutative ring spectrum below is the least under which cycle classes can be discussed: intersection of transverse cycles is a commutative product and the fundamental class of the ambient variety is a unit, so $E^*(X)$ must be a commutative ring.

Definition 3.1: Cycle-Receiving Structure

Let E be a homotopy-commutative ring spectrum. A *cycle-receiving structure* on E assigns to every smooth quasi-projective $X/\mathbb{C}$ and every smooth closed subvariety $Z \subseteq X$ of codimension i a class

$$[Z]_E \;\in\; E^{2i}_Z(X)$$

in the cohomology of X with supports in Z, such that

1. *(base change)* for every morphism $g\colon X' \to X$ of smooth varieties transverse to Z, $g^*[Z]_E = [g^{-1}Z]_E$ in $E^{2i}_{g^{-1}Z}(X')$, and $[\emptyset]_E = 0$.
2. *(multiplicativity and unit)* $[Z]_E \smile [W]_E = [Z \cap W]_E$ in $E^{2i+2j}_{Z\cap W}(X)$ whenever Z and W are smooth of codimensions i, j and meet transversally, and $[X]_E = 1$ in $E^0_X(X) = E^0(X)$.
3. *(comparison)* the image $\overline{[Z]}_E \in E^{2i}(X)$ lies in F^{2i}. The class $\mathrm{cl}_i([Z])$, viewed in $E_2^{2i,0} = H^{2i}(X; E^0)$ along the coefficient map induced by the unit $\mathbb{Z} = \pi_0\mathbb{S} \to E^0$, is a permanent cycle, and the class of $\overline{[Z]}_E$ in $E_\infty^{2i,0}$ is its class there.[a]

Extending additively over cycles supported on smooth subvarieties gives $\psi_X\colon Z^i(X)^{\mathrm{sm}} \to E^{2i}(X)$, $Z \mapsto \overline{[Z]}_E$, with $\varepsilon_X \circ \psi_X = \mathrm{cl}_i$ after the coefficient extension $\mathbb{Z} \to E^0$, where ε_X denotes the edge homomorphism of E on F^{2i}, the θ of section 1 when E is KU.

[a] Axiom (3) is the factorization condition $\beta \circ \alpha = \mathrm{cl}_i$ of the naive factorization, with the map of spectra β replaced by the edge homomorphism. Its filtration clause is automatic, as a class supported on a subvariety of complex codimension i restricts to zero on the $(2i-1)$-skeleton of a dual cell structure. Taking values in $E^{2i}_Z(X)$ rather than $E^{2i}(X)$ is the usual convention for a cycle class map, and is how Panin's orientations are defined at [36, Def. 3.1].

The definition axiomatises the minimal properties that KU and MU have, and theorem 3.2 shows that anything with those properties is oriented. Its contribution to the twisted case is that it makes orientation the only mechanism by which a cycle can be received, so any factorization *must* induce one. In particular it already excludes twisted K-theory with the pair $(\oplus, \otimes)$ of section 2: the product $\otimes$ is not well-defined in $KU^0_{[H]}(X)$ and has no unit there. The exclusion of twisted K-theory is proved independent of which product one attempts to use in section 3.2.

The first choice of definition would have been the literal triangle

$$\begin{tikzcd} & E^{2i}(X) \arrow[dr, "b"] & \\ \mathrm{CH}^i(X) \arrow[ur, "a"] \arrow[rr, "\mathrm{cl}_i"'] & & H^{2i}(X;\mathbb{Z}) \end{tikzcd}$$

with b induced by a map of spectra $E \to H\mathbb{Z}$ and a defined on all of $\mathrm{CH}^i(X)$. Two weakenings separate definition 3.1 from it. First, the comparison is the edge homomorphism rather than a map of spectra. A spectrum map would make b a map of cohomology theories, commuting with suspension and natural in X, but no unital map $KU \to H\mathbb{Z}$ exists: it would induce a graded ring map $KU_* \to \mathbb{Z}$, and as $\mathbb{Z}$ sits in degree zero the Bott class $\zeta \in KU_2$ and ζ^{-1} both map to 0, giving $1 = \zeta \cdot \zeta^{-1} \mapsto 0$. The comparison that KU and MU do carry, and the one Atiyah-Hirzebruch and Totaro use, is the edge homomorphism, which is already additive and natural in X.

Second, the domain is cycles supported on *smooth* subvarieties rather than $\mathrm{CH}^i(X)$. A singular Z must be resolved, $\xi\colon \widetilde{Z} \to Z \hookrightarrow X$, and $\xi_*(1)$ depends on the resolution. The smooth case suffices, as the orientation is extracted in section 3.1 from the classes of hyperplanes in $\mathbb{P}^n$, which are smooth. We do not claim that cycles with smooth support generate $\mathrm{CH}^i(X)$: the resolution dependence is removed instead by passing to $MU^{2i}(X) \otimes_{MU^*} \mathbb{Z}$, the quotient along the *augmentation* $MU^* \to \mathbb{Z}$ that kills every polynomial generator, which is why Totaro's refinement lands there and not in $MU^{2i}(X)$ [2].

Theorem 3.2: Cycle-Receiving Theorem

Let E be a homotopy-commutative ring spectrum carrying a cycle-receiving structure. Then

1. E is *complex-oriented*: there is a class $x \in \widetilde{E}^2(\mathbb{CP}^\infty)$ restricting to the unit of E^0 under $\widetilde{E}^2(\mathbb{CP}^1) \cong E^0$, equivalently a ring map $\varphi_x\colon MU \to E$.

2. The structure is the Gysin system of that orientation: for $\iota\colon Z \hookrightarrow X$ smooth closed of codimension i, the class $[Z]_E$ is the Thom class of $N_{Z/X}$ under $E^{2i}_Z(X) \cong \widetilde{E}^{2i}(\mathrm{Th}\, N_{Z/X})$, and
$$\overline{[Z]}_E = \iota_*(1) = \varphi_x\big([Z]_{MU}\big).$$

A theory that is not complex-oriented carries no cycle-receiving structure.

Assertion (2) states that a cycle-receiving structure *is* the Gysin system of an orientation, so no construction of a cycle class can evade the orientation formalism by proceeding differently. This is what rules out using the Gysin maps that twisted K-theory does possess (section 3.2). The proof occupies section 3.1 and has four steps. Step 0 transports E to a ring cohomology theory A^* on smooth $\mathbb{C}$-varieties, the setting in which the Panin-Smirnov correspondence is stated. Step 1 extracts the orientation from the values of the structure on projective spaces and converts it into φ_x by Quillen's theorem, proving (1). Step 2 turns the orientation into a system of pushforwards along projective morphisms. Step 3 shows that the Thom classes the structure defines by itself, on the zero sections of vector bundles, are those of that orientation, which gives (2). As a consequence, no complex-oriented theory refines cl_i further than MU does.

Remark 3.3 (Cycles that are differently oriented). Theorem 3.2 is a statement about definition 3.1, and it does not assert that a cycle class of every kind forces a complex orientation. The SL-oriented theories of Panin-Walter [39], for example Hermitian K-theory and the Witt-sheaf theories, carry Thom classes and pushforwards along a morphism only once the determinant of its normal bundle is trivialized. They therefore attach a class to the pair $\big(Z,\ \text{a trivialization of } \det N_{Z/X}\big)$ and not to Z alone, which is not the assignment $Z \mapsto [Z]_E$ that definition 3.1 requires, defined for every smooth closed Z and stable under every transverse base change, so no contradiction with theorem 3.2 arises. The added flexibility has content in a different problem: over $\mathbb{R}$ the determinant twist is exactly the extra information that lets such theories refine the cycle map to Chow-Witt groups, and they produce obstructions for the *real* integral Hodge conjecture of Benoist-Wittenberg [10], where algebraic cycles retain information about the real locus $X(\mathbb{R})$. The questions they detect concern $X(\mathbb{R})$. Over $\mathbb{C}$ they add nothing, namely the complex realization carries these theories and their Gysin classes into topological K-theory, so every condition they impose on the classes of section 1 is a K-theoretic condition.

3.1 A Cycle-Receiving Structure Must Be Oriented

Step 0: Algebraic vs. Analytic

The proof crosses between the algebraic and the analytic side, so the objects are fixed first. The groups $\mathrm{CH}^i(X)$ and the subvarieties $Z \subseteq X$ live on $\mathrm{Sm}_{\mathbb{C}}$, the category of smooth quasi-projective $\mathbb{C}$-varieties, while E is a spectrum, and orientations as stated in theorem 3.2 are for complex vector bundles on topological spaces. Analytification $X \mapsto X(\mathbb{C})$ is a functor $\mathrm{Sm}_{\mathbb{C}} \to \mathrm{Top}$ carrying a smooth closed subvariety of codimension i to a closed complex submanifold of real codimension $2i$, an algebraic vector bundle to a complex one, and a regular embedding to one whose normal bundle is complex. Set

$$A^p(X) := E^{2p}(X(\mathbb{C})), \qquad A^p_Z(X) := E^{2p}_{Z(\mathbb{C})}(X(\mathbb{C})), \qquad X \in \mathrm{Sm}_{\mathbb{C}},\ Z \subseteq X \text{ closed}$$

the doubling of degrees making codimension the grading, so that a codimension-i cycle class lies in $A^i(X)$. The odd-degree groups $E^{2p+1}(X(\mathbb{C}))$ remain part of the theory and enter through the boundary maps of the localization sequence. Only the even part carries the cycle classes, and Panin's formalism is independent to the grading convention.[4]

Then A^* satisfies Panin's axioms for a cohomology theory on $\mathrm{Sm}_{\mathbb{C}}$ [36, Def. 2.1], and is multiplicative because E is a ring spectrum, so it is a ring cohomology theory in his sense:

1. *Boundary and localization.* The long exact sequence of a topological pair supplies the connecting homomorphisms and localization sequence.
2. *Homotopy invariance.* The algebraic projection $(\mathbb{A}^1 \times X)(\mathbb{C}) \to X(\mathbb{C})$ induces a homotopy equivalence on complex points.
3. *Nisnevich excision.* An étale morphism $g\colon X' \to X$ inducing an isomorphism $g^{-1}(Z) \xrightarrow{\cong} Z$ is a local biholomorphism injective on $g^{-1}(Z)(\mathbb{C})$. It therefore restricts to an isomorphism between open analytic neighbourhoods of $g^{-1}(Z)(\mathbb{C})$ and $Z(\mathbb{C})$, so classical topological excision applies.

On the other hand, the classifying space $\mathbb{CP}^\infty = \varinjlim_n \mathbb{P}^n(\mathbb{C})$ is realized as a filtered colimit of analytifications of smooth projective varieties.

Overall, the algebraic-to-analytic passage allows the cycle classes on the $\mathbb{P}^n$ to construct the universal orientation of the topological spectrum E (Step 1), while the functor A^* provides the algebraic cohomology theory on $\mathrm{Sm}_{\mathbb{C}}$ to which the Panin-Smirnov correspondence applies (Step 2), converting that orientation into pushforwards along projective algebraic morphisms.

Step 1: The orientation and the ring map φ_x

Let E carry a cycle-receiving structure. We build from it a complex orientation, i.e. a class $x \in \widetilde{E}^2(\mathbb{CP}^\infty)$ whose restriction to $\widetilde{E}^2(\mathbb{CP}^1) \cong E^0$ is a unit. Let $H_n \subset \mathbb{P}^n$ be a hyperplane and set

$$x_n \ := \ \overline{[H_n]}_E \ \in \ E^2(\mathbb{P}^n)$$

the image of the supported class under the forgetful map $E^2_{H_n}(\mathbb{P}^n) \to E^2(\mathbb{P}^n)$, natural in $\mathbb{P}^n$. It is independent of the choice of hyperplane: any two hyperplanes of $\mathbb{P}^n$ are related by an element of the connected group $\mathrm{PGL}_{n+1}(\mathbb{C})$, which is homotopic to the identity and so acts trivially on $E^2(\mathbb{P}^n)$. Fixing a basepoint and choosing H_n to miss it, restriction along that point gives $x_n|_{\mathrm{pt}} = \overline{[\emptyset]}_E = 0$, so every x_n is reduced. Choose a linear embedding $j\colon \mathbb{P}^n \hookrightarrow \mathbb{P}^{n+1}$ and a hyperplane H_{n+1} not containing $j(\mathbb{P}^n)$. Then j is transverse to H_{n+1}, its preimage is a hyperplane of $\mathbb{P}^n$, and definition 3.1(1) together with naturality of the forgetful map gives $j^*(x_{n+1}) = x_n$. The Milnor sequence [1, section 3.5]

$$0 \longrightarrow \varprojlim_n{}^1 \widetilde{E}^1(\mathbb{P}^n) \longrightarrow \widetilde{E}^2(\mathbb{CP}^\infty) \longrightarrow \varprojlim_n \widetilde{E}^2(\mathbb{P}^n) \longrightarrow 0$$

[4] For 2-periodic E, such as KU, the resulting theory is $\mathbb{Z}/2\mathbb{Z}$-graded. The orientation formalism in that setting is developed in [32, section 1]. Periodicity is harmless here, as the codimension of a cycle is recorded by the filtration F^{2i} and not by the grading.

is surjective on the right, so some $x \in \widetilde{E}^2(\mathbb{CP}^\infty)$ restricts to the compatible system (x_n). The $\varprojlim^1$ term affects only the uniqueness of x, which is not used.[5]

It remains to check that x restricts to a generator on $\mathbb{CP}^1$. For $X = \mathbb{P}^1$ the skeletal filtration has a single nonzero step, $F^2E^2(\mathbb{P}^1) = \widetilde{E}^2(\mathbb{P}^1)$ and $F^3 = 0$, and the Atiyah-Hirzebruch spectral sequence collapses: the only nonzero columns are $p = 0, 2$, and the $p = 0$ column consists of permanent cycles because restriction to the basepoint is split by the projection to a point, so no differential acts. The edge homomorphism $\widetilde{E}^2(\mathbb{P}^1) \to E_\infty^{2,0} = H^2(\mathbb{P}^1; E^0) \cong E^0$ is then the suspension isomorphism. Since $\mathrm{cl}_1(H_1)$ generates $H^2(\mathbb{P}^1;\mathbb{Z})$, definition 3.1(3) says that x_1 corresponds to the unit of E^0. Thus x is a complex orientation.

The passage from that orientation to a ring map $MU \to E$ is Quillen's theorem. The class x defines a *first Chern class* $c_1^E(L) := f_L^*(x) \in E^2(Y)$ for every complex line bundle L on a space Y, where $f_L\colon Y \to \mathbb{CP}^\infty$ is the classifying map. On $\mathbb{CP}^\infty$ it gives $E^*(\mathbb{CP}^\infty) = E^*(\mathrm{pt})[[x]] = E^*[[x]]$ and $E^*(\mathbb{CP}^\infty \times \mathbb{CP}^\infty) = E^*[[u, v]]$, so the first Chern class of the external tensor product of the two tautological line bundles is a power series

$$c_1^E(L \otimes M) = F_E\big(c_1^E(L), c_1^E(M)\big), \qquad F_E \in E^*[[u, v]]$$

The identities $\mathcal{O} \otimes L \cong L$, commutativity, and associativity of $\otimes$ make F_E a one-dimensional commutative formal group law over E^*. The Lazard ring corepresents formal group laws[6], and $\mathbb{L}^* \cong \mathbb{Z}[t_1, t_2, \dots]$ with $|t_i| = 2i$ [29], [3, Part II], so F_E is classified by a ring map $\mathbb{L}^* \to E^*$. Then A complex orientation of E is equivalent to a homotopy ring map $MU \to E$ as such a map pulls the tautological orientation of MU back to one of E, and conversely an orientation determines the map on the Thom spectrum [3, Part II]. Quillen's theorem [41] identifies MU^* with $\mathbb{L}^*$ so that the formal group law of the tautological orientation is the universal one, and under the correspondence the map $\varphi_x\colon MU \to E$ attached to x classifies F_E on coefficients. Producing x and φ_x establishes assertion (1) of theorem 3.2.

Two consequences are recorded for later use. Writing φ_x also for the induced maps $MU^*(Y) \to E^*(Y)$ and $MU^* \to E^*$ on coefficients, one has

$$c_1^E(L) = \varphi_x\big(c_1^{MU}(L)\big) \text{ for every line bundle } L, \qquad \text{and hence} \qquad F_E = \varphi_x\big(F_{MU}\big)$$

the second identity meaning that the coefficient map $\varphi_x\colon MU^* \to E^*$ applied to the coefficients of the power series F_{MU} gives F_E. The first identity holds because φ_x is a ring map carrying the tautological orientation x^{MU} to x, so $\varphi_x\left(f_L^*(x^{MU})\right) = f_L^*\left(\varphi_x(x^{MU})\right) = f_L^*(x)$, and the second follows by applying it to $L \otimes M$ on $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$. In this sense φ_x carries the orientation of MU to that of E.

Step 2: from the orientation to pushforwards

Assertion (2) of theorem 3.2 refers to the Gysin maps, so the orientation must first be converted into a system of pushforwards $f_*\colon A^*(Y) \to A^{*+d}(X)$ along projective morphisms f of relative codimension d. This is the step that requires the algebraic build-up from step 0: the cycles to be pushed forward are the closed immersions $\iota\colon Z \hookrightarrow X$ of a codimension-i subvariety, which are projective morphisms, and a projective morphism factors [20, p. II.4] as

$$f\colon Y \xrightarrow{\iota} \mathbb{P}_X^N \xrightarrow{\pi} X$$

[5] Once (1) is proved, $E^*(\mathbb{P}^n)$ is a free E^*-module on $1, x, \dots, x^n$ and the restriction maps $\widetilde{E}^*(\mathbb{P}^{n+1}) \to \widetilde{E}^*(\mathbb{P}^n)$ are surjective, so the tower is Mittag-Leffler, the $\varprojlim^1$ term vanishes, and x is in fact unique. Note the system (x_n) is the one determined by the given cycle-receiving structure, and a different structure on the same E determines a different system, hence a different orientation (remark 3.6).

[6] corepresents means that if $\mathrm{FGL}\colon \mathbf{CRing} \to \mathbf{Set}$ assigns to each ring R the set of 1-dimensional commutative formal group laws over R, then $\mathrm{FGL}(-) \cong \mathrm{Hom}(\mathbb{L}^*, -)$

with ι a closed immersion and π the structural projection. A system of pushforwards is therefore determined by its values on regular embeddings, computed from the Thom isomorphism of the normal bundle, and on projective-bundle projections, computed from the projective bundle formula. Note that ι_* raises degree by i because it is the Thom isomorphism of the rank-i normal bundle followed by extension of supports, placing $\iota_*(1)$ in $A^i(X)$.[7] That both are in fact determined by the first Chern class of line bundles alone, and that the resulting system exists and is unique, is the theorem of Panin and Smirnov recalled below.

Note that the bridge of Step 0 gives A^*, built from the topological spectrum E by analytification. In the other direction, the orientation x of Step 1 lives on the topological side, and the assignment displayed next pulls x back to a class in A^1. Everything Panin-Smirnov then produces, in particular the pushforward ι_* along an algebraic closed immersion, is a construction in A^*.

Transport the orientation to A^* by

$$c_1(L) := f_{L^{\mathrm{an}}}^*(x) \;\in\; E^2(X(\mathbb{C})) = A^1(X)$$

where L is an algebraic line bundle on $X \in \mathrm{Sm}_{\mathbb{C}}$, L^{an} is its analytification, a topological complex line bundle on $X(\mathbb{C})$, and $f_{L^{\mathrm{an}}}\colon X(\mathbb{C}) \to \mathbb{CP}^\infty$ is the classifying map of L^{an}. This assignment is natural in X, vanishes on $\mathcal{O}_X$ because $f_{\mathcal{O}^{\mathrm{an}}}$ is constant, and is nondegenerate in the sense that $\big(1, c_1(\mathcal{O}_{\mathbb{P}^1}(-1))\big)$ is an $A^*(X)$-basis of $A^*(X \times \mathbb{P}^1)$, since x is an orientation and the projective bundle theorem holds for complex-oriented theories [3, Part II]. These three properties are exactly Panin's definition of a *Chern structure* on A^* [36, Def. 3.2]: an assignment

$$\big(L \to X\big) \longmapsto c_1(L) \in A^1(X), \qquad X \in \mathrm{Sm}_k$$

that is natural, vanishing on the trivial bundle, and nondegenerate.

Panin and Smirnov show that a Chern structure imposed on line bundles determines a unique system of pushforwards along all projective morphisms. For a ring cohomology theory A^* on Sm_k they consider four structures:

1. an *orientation*: a rule assigning to each X, each closed $Z \subseteq X$ and each vector bundle $V \to X$ a grade-preserving two-sided $A^*(X)$-module isomorphism (a *Thom isomorphism*) $\omega_Z^V\colon A_Z^*(X) \to A_Z^*(V)$, invariant under bundle isomorphisms, compatible with base change, and multiplicative for direct sums [37, Def. 1.9]
2. a *Thom structure*: an assignment of classes $\mathrm{th}(L) \in A_X^1(L)$ to line bundles, natural, and normalised so that cup product with $\mathrm{th}(L)$ is an isomorphism [36, Def. 3.3]
3. a *Chern structure*, as above [36, Def. 3.2]
4. an *integration* (trace structure): operators $f_*\colon A^*(Y) \to A^{*+d}(X)$ for every projective f of relative codimension d, two-sided $A^*(X)$-linear (the projection formula), functorial, compatible with transverse base change and with the projections $\mathbb{P}^n \times Y \to Y$, normalised by $(\mathrm{id}_X)_* = \mathrm{id}$, and satisfying localization, i.e. exactness of the Gysin sequence $A^*(Z) \to A^*(X) \to A^*(X \setminus Z)$ for a closed immersion $Z \hookrightarrow X$ [37, Def. 2.2]

Their theorems [36, Thms. 3.5, 3.35, 3.36] and [37, Thm. 2.5] assemble these into a commuting square in which every arrow is a bijection:

$$\begin{tikzcd}
\{\text{orientations on } A^*\} \arrow[r, "\alpha"] \arrow[d, "\epsilon"'] & \{\text{integrations on } A^*\} \arrow[d, "\beta"] \\
\{\text{Thom structures on } A^*\} \arrow[r, "\gamma"'] & \{\text{Chern structures on } A^*\} \arrow[ul, dashed, "\delta"]
\end{tikzcd}$$

[7] In section 6 the projective resolutions $\xi\colon \widetilde{Z} \to X$ of singular cycles are also needed. These are likewise projective, so the reduction works in both the algebraic and the analytic settings.

with $\delta := (\gamma\epsilon)^{-1}$, so that a structure of any one of the four kinds determines the other three, and $\beta\alpha = \gamma\epsilon$. The correspondence between Chern structure and integrations is explicit:

- *Integration to Chern structure (β):* The zero section $z\colon X \hookrightarrow L$ of a line bundle gives the Thom class $\mathrm{th}(L) := z_*(1)$ and first Chern class $c_1(L) := z^* z_*(1) \in A^1(X)$
- *Chern structure to Thom structure (γ^{-1}):* The projective bundle formula

$$A^*(\mathbb{P}(V)) = \bigoplus_{j=0}^{r-1} A^*(X) \cdot c_1\big(\mathcal{O}_{\mathbb{P}(V)}(-1)\big)^j, \qquad r = \operatorname{rk} V$$

  propagates the Chern structure to higher ranks via the splitting principle, and the top Chern class of $\mathcal{O}_{\mathbb{P}(1\oplus V)}(1) \otimes p^*V$ constructs the Thom class $\mathrm{th}(V)$, giving the Thom isomorphisms $A^*(X) \xrightarrow{\sim} A_X^{*+\mathrm{rk}V}(V)$
- *Chern structure to Integration ($\beta^{-1} = \alpha\delta$):* Deformation to the normal cone identifies $A_Z^*(X)$ with $A_Z^*(N_{Z/X})$ to produce Gysin maps ι_* for regular embeddings. Then the projective bundle formula produces π_* for smooth projections, and Panin's theorem is that setting $f_* = \pi_* \circ \iota_*$ for any projective factorization gives a well-defined integration, the *unique* one satisfying $\iota_*(1) = c_1(L(D))$ for smooth divisors $\iota\colon D \hookrightarrow X$ [37, Thm. 2.5]

Consequently A^* carries a unique integration inducing c_1, and with it a Gysin class $\iota_*(1) \in A^i(X)$ for every smooth closed $Z \subseteq X$ of codimension i. Applying the same construction to $MU^{2*}(-(\mathbb{C}))$ with its tautological orientation, and writing $[Z]_{MU} := \iota_*^{MU}(1)$, one has

$$\varphi_x\big([Z]_{MU}\big) = \iota_*(1). \tag{3}$$

Indeed ι_* is, in both theories, the composite of the Pontryagin-Thom collapse $X_+ \to \operatorname{Th} N_{Z/X}$ with the Thom isomorphism: the collapse is a map of spaces, so φ_x commutes with it, and φ_x carries the Thom class of MU to that of E because it is a ring map carrying c_1^{MU} to c_1^E and both Thom classes are built from c_1 by the construction just recalled. In the algebraic setting this is the degenerate case of Panin's Riemann-Roch theorem [38]: for a morphism φ of ring cohomology theories, each carrying an orientation, and a closed embedding $\iota\colon Z \hookrightarrow X$ of smooth varieties,

$$\varphi\big(\iota_*(a)\big) = \iota_*\big(\varphi(a) \cdot \mathrm{itd}_\varphi(N_{Z/X})\big)$$

where itd_φ is the inverse Todd class of φ,[8] and since φ_x carries c_1^{MU} to c_1^E exactly, its Todd class is 1 and the formula reduces to the commutation (3). Uniqueness of the integration is part of [37, Thm. 2.5].

Step 3: the structure is that Gysin system

There are now two classes attached to a smooth closed $Z \subseteq X$: the class $[Z]_E \in E_Z^{2i}(X)$ given by the cycle-receiving structure, and the Gysin class $\iota_*(1) = \varphi_x([Z]_{MU})$ built from the orientation. We show they agree.

Let $V \to Y$ be an algebraic vector bundle of rank r over a smooth quasi-projective Y. Its total space is smooth quasi-projective and its zero section $z\colon Y \hookrightarrow V$ has image a smooth closed subvariety of codimension r, so the structure gives a class

$$\mathrm{th}(V) \;:=\; [\,0_V\,]_E \;\in\; E_Y^{2r}(V) \;\cong\; \widetilde{E}^{2r}(\operatorname{Th} V)$$

[8] itd_φ is the multiplicative inverse of the Todd class td_φ, the characteristic class measuring the failure of φ to carry the orientation of its source to that of its target.

where $\mathrm{th}(V)$ is the Thom class, and the identification is by excision and the tubular neighbourhood theorem. The assignment $V \mapsto \mathrm{th}(V)$ is a system of Thom classes in Panin's sense [37, Def. 1.10], that is, it satisfies:

- *Naturality and invariance.* For $h\colon Y' \to Y$ the map $h^*(V) \to V$ is transverse to 0_V with preimage 0_{h^*V}, so axiom (1) gives $\mathrm{th}(h^*V) = h^*\mathrm{th}(V)$. The same axiom applied to an isomorphism $V \cong W$ of bundles over Y gives $\mathrm{th}(V) = \mathrm{th}(W)$ under it.
- *Multiplicativity.* The projections π_V, π_W of $V \oplus W$ are smooth, hence transverse to the zero sections, so $[V \oplus 0_W]_E = \pi_W^*\mathrm{th}(W)$ and $[0_V \oplus W]_E = \pi_V^*\mathrm{th}(V)$. These two subvarieties meet transversally in $0_{V\oplus W}$, so axiom (2) gives $\mathrm{th}(V \oplus W) = \mathrm{th}(V) \smile \mathrm{th}(W)$ under $\mathrm{Th}(V \oplus W) = \mathrm{Th}\, V \wedge \mathrm{Th}\, W$.
- *The Thom isomorphism.* For the trivial line bundle it suffices to compute over $Y = \mathrm{pt}$. The open immersion $j\colon \mathbb{A}^1 \hookrightarrow \mathbb{P}^1$ is transverse to $\{0\}$ with preimage $\{0\}$, so axiom (1) gives $j^*[\,\{0\} \subset \mathbb{P}^1\,]_E = \mathrm{th}(\mathcal{O})$, and on supported groups j^* is the excision isomorphism $E^2_0(\mathbb{P}^1) \cong E^2_0(\mathbb{A}^1)$. Forgetting supports sends $[\,\{0\} \subset \mathbb{P}^1\,]_E$ to the hyperplane class x_1 of Step 1, a point of $\mathbb{P}^1$ being a hyperplane, so under $E^2_0(\mathbb{A}^1) \cong \widetilde{E}^2(\mathbb{P}^1) \cong E^0$ the class $\mathrm{th}(\mathcal{O})$ corresponds to x_1, the unit by the normalisation of Step 1: $\mathrm{th}(\mathcal{O})$ is the suspension generator. Restricting $\mathrm{th}(V)$ to a fibre therefore gives $\mathrm{th}(\mathbb{C}^r) = \mathrm{th}(\mathcal{O})^{\smile r}$, the generator of $\widetilde{E}^{2r}(S^{2r})$ over E^0 by the previous two properties, so cup product with $\mathrm{th}(V)$ is an isomorphism $E^*(Y) \xrightarrow{\sim} \widetilde{E}^{*+2r}(\mathrm{Th}\, V)$ by Mayer-Vietoris over a finite trivializing cover. This is Panin's normalisation axiom, and it is the *only* place where the comparison axiom (3) enters Step 3.

Hence th is an orientation of A^* in the sense of [37, Def. 1.9, 1.10], and its restriction to line bundles is a Thom structure. Its Chern structure is the c_1 transported in Step 2. Indeed, the continuous sections of a line bundle $L \to Y$ form a vector space, so any algebraic section is homotopic to the zero section through continuous sections. As E^* is homotopy invariant, and A^* with it, the two induce the same map on A^*. If moreover s is transverse to 0_L then axiom (1) applies to s and

$$c_1^{\mathrm{th}}(L) \;:=\; z^*\overline{\mathrm{th}(L)} = s^*\overline{\mathrm{th}(L)} = \overline{[\,s^{-1}(0)\,]}_E$$

For $L = \mathcal{O}(1)$ on $\mathbb{P}^n$ a linear form is such a section and cuts out a hyperplane, so

$$c_1^{\mathrm{th}}(\mathcal{O}(1)) = \overline{[H_n]}_E = x_n = c_1(\mathcal{O}(1))$$

A very ample L is $j^*\mathcal{O}(1)$ for an immersion $j\colon Y \to \mathbb{P}^N$, so naturality gives $c_1^{\mathrm{th}}(L) = c_1(L)$. Then the two structures carry the same formal group law, since both read it off $c_1(\mathcal{O}(1) \boxtimes \mathcal{O}(1))$ on $\mathbb{P}^n \times \mathbb{P}^m$ (the Segre pullback of $\mathcal{O}(1)$), and as every line bundle on a smooth quasi-projective variety is $L_1 \otimes L_2^\vee$ with L_i very ample [20, p. II.7], $c_1^{\mathrm{th}} = c_1$ on all line bundles.

By the bijection between orientations and Chern structures, two orientations with the same Chern structure coincide, so $\mathrm{th} = \mathrm{th}^x$ for th^x the orientation determined by x.

It remains to identify the class of an arbitrary smooth closed $Z \subseteq X$ with the Thom class of its normal bundle. The tubular neighbourhood of $Z(\mathbb{C})$ in $X(\mathbb{C})$ is not algebraic, so axiom (1) cannot be applied to it directly. The comparison instead passes through the deformation to the normal bundle, which is algebraic.

Lemma 3.4: Deformation to the Normal Bundle

Let E carry a cycle-receiving structure, let $Z \subseteq X$ be a smooth closed subvariety of codimension i, and let $N = N_{Z/X}$. Under the identification $E^{2i}_Z(X) \cong E^{2i}_{0_N}(N)$ furnished by the tubular neighbourhood theorem and excision,

$$[\,Z \subset X\,]_E \;=\; [\,0_N \subset N\,]_E \;=\; \mathrm{th}(N).$$

Proof **:**
Let M° be the deformation space: the blow-up of $X \times \mathbb{A}^1$ along $Z \times \{0\}$, with the strict transform of $X \times \{0\}$ removed [35, section 5.1]. It is smooth and quasi-projective, its fibres over $\mathbb{A}^1$ are $M^\circ_t = X$ for $t \neq 0$ and $M^\circ_0 = N$, and the strict transform W of $Z \times \mathbb{A}^1$ is a smooth closed subvariety of codimension i, isomorphic to $Z \times \mathbb{A}^1$ because the centre $Z \times \{0\}$ is a Cartier divisor in $Z \times \mathbb{A}^1$, meeting the fibres in $W_1 = Z$ and $W_0 = 0_N$.

The fibre inclusions $j_1\colon X \hookrightarrow M^\circ$ and $j_0\colon N \hookrightarrow M^\circ$ are transverse to W: the tangent space of W contains the $\mathbb{A}^1$-direction, which supplements the tangent space of a fibre. Axiom (1) therefore gives

$$j_1^*\,[W]_E = [Z]_E \;\text{ in } E^{2i}_Z(X), \qquad j_0^*\,[W]_E = [0_N]_E = \mathrm{th}(N) \;\text{ in } E^{2i}_{0_N}(N).$$

Topologically, a complex vector bundle over $W(\mathbb{C}) = Z(\mathbb{C}) \times \mathbb{C}$ is pulled back from $Z(\mathbb{C})$ along the homotopy equivalence $\mathrm{pr}\colon Z(\mathbb{C}) \times \mathbb{C} \to Z(\mathbb{C})$, so the normal bundle of $W(\mathbb{C})$ in $M^\circ(\mathbb{C})$ is pr^*N. Excision and the tubular neighbourhood theorem therefore identify all three supported groups with $\widetilde{E}^{2i}(\mathrm{Th}\,N)$, and under these identifications j_1^* and j_0^* are both inverse to the isomorphism pr^*, hence agree (that the two restrictions are isomorphisms is, for the theories of Step 0, also the statement of [37, Thm. 1.2]). The two displayed restrictions of $[W]_E$ therefore correspond, and the composite identification $E^{2i}_Z(X) \cong E^{2i}_{0_N}(N)$ is the tubular one, completing the proof.

With the lemma, for $\iota\colon Z \hookrightarrow X$ smooth closed of codimension i,

$$[Z]_E = \mathrm{th}(N_{Z/X}) = \mathrm{th}^x(N_{Z/X})$$

The Gysin map is by definition the composite $E^0(Z) \cong E^{2i}_Z(X) \to E^{2i}(X)$ formed from the Thom isomorphism of th^x, so with (3),

$$\overline{[Z]}_E = \iota_*(1) = \varphi_x([Z]_{MU})$$

completing the proof of theorem 3.2.

Remark 3.5 (Supports are needed only for the second assertion). Suppose one weakens definition 3.1 by keeping only the images $\overline{[Z]}_E \in E^{2i}(X)$, with axioms (1) and (2) in unsupported form. Then (1) of theorem 3.2 is unaffected, since Step 1 uses only the hyperplane classes x_n in $E^2(\mathbb{P}^n)$, their compatibility under transverse linear embeddings, and the normalisation on $\mathbb{P}^1$. Assertion (2), however, becomes unprovable: without a class in $E^{2i}_Z(X)$ there is no Thom class to compare.

Remark 3.6 (Rigidity is relative to the induced orientation). Theorem 3.2(2) does not assert uniqueness of the structure, and indeed it is *not* unique. If x is an orientation and $f(t) = t + a_1 t^2 + \cdots$ with $a_i \in E^{-2i}$, then $f(x)$ is another orientation whose class on $\mathbb{P}^1$ is unchanged,[9] so its Gysin system is a second cycle-receiving structure. What the theorem says is that every such structure is the Gysin system of the orientation it *itself* induces, so a cycle class cannot be produced without an orientation.

[9] The reduced diagonal $\mathbb{P}^1 \to \mathbb{P}^1 \wedge \mathbb{P}^1$ is a map $S^2 \to S^4$, hence null-homotopic, so every product of reduced classes vanishes on $\mathbb{P}^1$, in particular $x^k|_{\mathbb{P}^1} = 0$ for $k \geq 2$ and $f(x)|_{\mathbb{P}^1} = x|_{\mathbb{P}^1}$. Note that $\widetilde{E}^4(\mathbb{P}^1) = \pi_{-2}E$ need not vanish, so it is the null-homotopy of the reduced diagonal, and *not* the vanishing of the group, that is used.

3.2 Twisted K-theory Receives No Cycle

As shown in section 2, the product of an $[H]$-twisted bundle with an $[H']$-twisted bundle is $[H+H']$-twisted:

$$KU^0_{[H]}(X) \otimes KU^0_{[H']}(X) \longrightarrow KU^0_{[H+H']}(X)$$

the $KU^0(X)$-module structure being the $[H']=0$ case. For $[H]\neq 0$ the theory $KU_{[H]}$ is an invertible KU-module and not a KU-algebra: it is the sections of a bundle of KU-modules each fibre of which is equivalent to KU, and invertibility means that the fibrewise smash product over KU of the bundles for $[H]$ and for $-[H]$ is the trivial bundle. This section shows that no multiplication whatsoever repairs this.

By definition 3.1 the pair $(KU_{[H]}, \oplus, \otimes)$ is not a valid input: axiom (2) requires $[Z]_E \smile [W]_E$ to lie in the group containing $[Z\cap W]_E$, whereas the two sides lie over the twists $2[H]$ and $[H]$, distinct summands of the graded ring $K^0_\bullet(X)$ of (2). The normalisation is equally unavailable: in a Γ-graded ring (a ring with a decomposition $R=\bigoplus_{\gamma\in\Gamma} R_\gamma$ over an abelian group Γ with $R_\gamma R_{\gamma'} \subseteq R_{\gamma+\gamma'}$, here $\Gamma = H^3(X;\mathbb{Z})$) the identity is homogeneous of degree 0,[10] and the degree-zero summand of $K^0_\bullet(X)$ is the untwisted $K^0(X)$, whose identity is the class of the trivial line bundle. Theorem 3.2 therefore does not apply to $(KU_{[H]}, \oplus, \otimes)$. What the theorem supplies is a specification of what any replacement must provide: a system of pushforwards together with a distinguished class to push.

By [13, Thm. 1.1] the pushforwards *exist*. For a differentiable map $f\colon Z\to X$ of compact manifolds and $\sigma \in H^3(X;\mathbb{Z})$ there is a map

$$f_!\colon KU^*_{f^*\sigma+W_3(f)}(Z) \longrightarrow KU^{*+d(f)}_{\sigma}(X)$$

where $W_3(f)$ is the image of $w_2(Z) - f^*w_2(X)$ under the Bockstein $H^2(-;\mathbb{Z}/2)\to H^3(-;\mathbb{Z})$, vanishing exactly when f is K-oriented,[11] and $d(f) = \dim Z - \dim X$ modulo 2. For a closed immersion $\iota\colon Z\hookrightarrow X$ of smooth $\mathbb{C}$-varieties of codimension c the normal bundle $N_{Z/X}$ is complex so $W_3(\iota)=0$ and the map reads

$$\iota_!\colon KU^0_{\iota^*[H]}(Z) \longrightarrow KU^{2c}_{[H]}(X) = KU^0_{[H]}(X)$$

the last identification by Bott periodicity. It is therefore not *a priori* clear that no normalisation with respect to $[H]$ produces the missing distinguished class. The following shows that none does, and its proof uses no hypothesis on the multiplication:

Proposition 3.7: Twisted K-theory Receives No Cycle

Let X be smooth quasi-projective over $\mathbb{C}$ and $[H]\in H^3(X;\mathbb{Z})$ with $[H]\neq 0$. Then the unit $1 \in H^0(X;\mathbb{Z}) = E_2^{0,0}$ of the $[H]$-twisted spectral sequence is not a permanent cycle, so no class of $KU^0_{[H]}(X)$ satisfies the codimension-zero instance of definition 3.1(3) for $Z = X$.

Consequently, whatever homotopy-commutative multiplication is imposed on the twisted groups, they carry no cycle-receiving structure, and no factorization $\mathrm{CH}^i(X)\to KU^0_{[H]}(X)\to H^{2i}(X;\mathbb{Z})$ of the form of definition 3.1 exists.

[10] If $1=\sum_\gamma u_\gamma$ then for homogeneous a of degree δ the identity $a = 1\cdot a$ compares componentwise to $a = u_0 a$ and $u_\gamma a = 0$ for $\gamma\neq 0$, hence $1-u_0$ annihilates every homogeneous element and $1-u_0 = (1-u_0)\cdot 1 = 0$.

[11] f is *K-oriented* when its stable normal bundle carries a Spin^c-structure. This is the condition under which the Thom isomorphism of that bundle exists in K-theory, and it is the K-theoretic instance of the orientability discussed in section 3.1.

***Proof* :**
A cycle-receiving structure applied to $Z = X$ in codimension zero produces a class $[X]_E$ whose symbol in $E_\infty^{0,0}$ axiom (3) requires to be that of $\mathrm{cl}_0([X]) = 1$. In particular 1 must be a permanent cycle of the spectral sequence of the theory on X. That sequence is the $[H]$-twisted one of section 2: its E_2-page is $H^p(X; KU^q(\mathrm{pt}))$, because the twist is pulled back from the 2-connected $K(\mathbb{Z},3)$, the class in question is $1 \in H^0(X;\mathbb{Z}) = E_2^{0,0}$, and the coefficient map $\mathbb{Z} = \pi_0\mathbb{S} \to KU^0(\mathrm{pt}) = \mathbb{Z}$ is the identity. The first differential is

$$d_3^{[H]}(1) = \mathrm{Sq}^3_{\mathbb{Z}}(1) + [H] \smile 1 = [H],$$

since the Steenrod operations annihilate the unit. No differential enters the corner $(0,0)$, its source sitting in negative columns, so 1 passes to $E_\infty^{0,0}$ exactly when every differential kills it, and $d_3^{[H]}(1) = [H] \neq 0$ already fails. The argument used neither the multiplication nor any axiom besides (3), so it applies to every candidate structure, completing the proof.

Thus, the obstruction is not the absence of a pushforward but the absence of anything *canonical* to push forward, namely what fails is the existence of a class with the properties of the fundamental class of X. The same gap appears one step earlier if the full twisted Gysin system of [13] is assumed. Definition 3.1(2) fixes the fundamental class of Z as the unit of $E^0(Z)$, so $[Z]_{KU_{[H]}}$ would have to be $\iota_!(1)$ with $1 \in KU^0_{\iota^*[H]}(Z)$. A rank-one element of that group is an $\iota^*[H]$-twisted line bundle, which exists only if $\iota^*[H] = 0$, and even then it is not canonical: any two choices differ by an element of $\mathrm{Pic}\,(Z)$. Section 7.2 represents this failure quantitatively: the image of the degree-zero edge map is $\mathrm{ind}(\alpha) \cdot \mathbb{Z}$, and it contains 1 only when the Brauer class, hence the twist, is trivial. What twisted K-theory records is whether the Brauer class splits on a subvariety, not which classes are algebraic. Section 7, and in particular the discussion after proposition 7.7, makes this precise.

The multiplicative axioms are thus what pin the unit: axiom (2) forces $[X]_E = 1$, and axiom (3) in codimension zero kills it. Keeping only additivity and the comparison in a single positive codimension i asks instead for an "additive" lift, which the Gysin system above does supply, and whether such a lift can exist is the subject of section 4. A different weakening, in which axiom (2) is made homogeneous over the group of twists and the object pushed forward is a twisted bundle rather than a unit, is the setting of the twisted conjecture and is carried out in section 7.1.

4 The Twist as a Standalone Obstruction

Though the cycle map cannot factor through a twisted theory, the twisted differential can still obstruct algebraicity on its own: it is an operation on integral cohomology, defined by the twisted spectral sequence whether or not any cycle lifts. Dropping the multiplicativity of definition 3.1 and keeping only additivity and the comparison gives an *additive lift*, which the twisted Gysin system can in principle supply. This section treats complex cobordism MU, whose primary differential the twist cannot move, and K-theory KU, whose primary differential it does move.

4.1 Twisting Cobordism and Hertl's Theorem

The remaining candidate is a twist of the *universal* complex-oriented theory MU rather than of KU. Twisting a ring spectrum means here what it meant for KU in section 2, and is made precise in definition 4.1. Mirroring the proof of proposition 3.7, a cycle-receiving structure on a twisted theory requires the unit of $E_2^{0,0}$ to be a permanent cycle, and its first differential is the degree-three characteristic class $w_3(\tau)$ of the twist computed in theorem 4.3 below, which for MU vanishes identically by Hertl's theorem. The codimension-zero test that killed $KU_{[H]}$ is therefore passed vacuously by a twisted MU, and the question shifts to whether the twist deforms the differentials at all: a twisted MU could improve on MU only by moving the primary differential away from $\mathrm{Sq}^3_{\mathbb{Z}}$. What follows shows that it cannot, so that even as a standalone obstruction MU_τ sees nothing that untwisted MU does not.

Let E be a multiplicative cohomology theory.[12] A twist of E is a map $\tau : X \to B\mathrm{GL}_1(E)$ to the classifying space of its units, equivalently a bundle of invertible E-modules over X. Its twisted cohomology is the homotopy of the spectrum of sections, and the twists governed by H^3 are those pulled back from a universal $f\colon K(\mathbb{Z},3) \to B\mathrm{GL}_1(E)$ [4].

Definition 4.1: Twisted Complex Cobordism

Let $[H] \in H^3(X;\mathbb{Z})$ be classified by $h\colon X \to K(\mathbb{Z},3)$ and let $f\colon K(\mathbb{Z},3) \to B\mathrm{GL}_1(MU)$ be a map. The associated twist is $\tau = f \circ h\colon X \to B\mathrm{GL}_1(MU)$, and the *$\tau$-twisted complex cobordism* $MU^*_\tau(X)$ is the cohomology represented by the bundle of MU-module spectra it classifies.

Like twisted K-theory it carries an Atiyah-Hirzebruch spectral sequence with $E_2 = H^*(X; MU^*)$ converging to $MU^*_\tau(X)$, whose differential $d_3^{MU,\tau}$ the twist may deform. A twisted MU could still supply a new obstruction through this differential, if some choice of f moved $d_3^{MU,\tau}$ away from the untwisted $\mathrm{Sq}^3_{\mathbb{Z}}$. That is excluded by a theorem of Hertl [22], from which theorem 4.3 draws the consequence for the obstruction.

Proposition 4.2: Twists of Complex Cobordism Are Ghosts (Hertl)

Every $H^3(-;\mathbb{Z})$-twist[a] of MU is a *ghost*: for every map $f\colon K(\mathbb{Z},3) \to B\mathrm{GL}_1(MU)$ the looped map Ωf is null-homotopic, so f induces the zero map on all positive homotopy groups [22, Thm. B]. Moreover the canonical twist $\zeta\colon K(\mathbb{Z},3) \to B\mathrm{GL}_1(M\mathrm{Spin}^c)$, and hence the H^3-twist of K-theory, does not factor through $B\mathrm{GL}_1(MU)$ [22, Thm. A].

[a] Any element of $H^3(-;\mathbb{Z})$, not only a torsion one.

***Proof* : Idea**
This is a summary of Hertl's argument.

(i) From twists to ring maps. A twist $f\colon K(\mathbb{Z},3) \to B\mathrm{GL}_1(MU)$ loops to an H-map $\Omega f\colon \mathbb{CP}^\infty \to \mathrm{GL}_1(MU)$, an assignment of units to line bundles, and the adjoint of Ωf is a homotopy ring map $\phi\colon \Sigma^\infty_+\mathbb{CP}^\infty \to MU$ out of the spherical group ring of $\mathbb{CP}^\infty$. Proving Ωf null-homotopic amounts to proving that ϕ vanishes on the reduced summand.

(ii) Reduction to homology. The Adams-Novikov spectral sequence computing $[\Sigma^\infty_+\mathbb{CP}^\infty, MU]$ collapses, because $MU_*(\mathbb{CP}^\infty)$ is a free MU_*-module with induced comodule structure, so ϕ is detected by the map it induces on MU-homology.

(iii) The coefficient computation. In $MU_2(MU) = MU_2\{1\} \oplus MU_0\{b_1\}$, with $\beta_1 \in MU_2(\mathbb{CP}^\infty)$ the standard generator and b_1 the first Hopf-algebroid generator, write $\phi_*(\beta_1) = \lambda_2 \cdot 1 + \lambda_0\, b_1$ with $\lambda_2 \in MU_2$ and $\lambda_0 \in \mathbb{Z}$. As a ring map, ϕ converts the Pontryagin product of $MU_*(\mathbb{CP}^\infty)$, the one induced by the H-space structure of $\mathbb{CP}^\infty$, into the product of $MU_*(MU)$, and the

[12] Strictly speaking, for the classifying space of units $B\mathrm{GL}_1(E)$ and its unit spectrum $\mathrm{gl}_1(E)$ to be well-defined as deloopings, E must carry the structure of an E_∞-ring spectrum (or an A_∞-ring spectrum for $B\mathrm{GL}_1(E)$ as a space). This guarantees that multiplication on E is associative and commutative up to coherent higher homotopies, allowing the space of units $\mathrm{GL}_1(E) \subset \Omega^\infty E$ to be delooped. Both KU and MU carry E_∞-structures, so the requirement is automatically satisfied and does not alter the Atiyah-Hirzebruch spectral sequence computations.

relation $\beta_1^{\bullet n} = n!\,\beta_n + \cdots$ forces the divisibilities $n!\big|\lambda_0^n$ for all n, hence $\lambda_0 = 0$. A comparison of coactions after mapping to K-theory kills λ_2. So ϕ vanishes on reduced homology, and with (ii) the map Ωf is null, which is Theorem B.

(iv) The formal-group reformulation (Thm. D). A homotopy ring map $\phi\colon \Sigma^\infty_+\mathbb{CP}^\infty \to MU$ restricts to a class $u \in MU^0(\mathbb{CP}^\infty)$ that converts the H-space sum of $\mathbb{CP}^\infty$ into the product: as a power series in the orientation variable, $u(F_{MU}(s,t)) = u(s)\,u(t)$. Such a u is a homomorphism from the universal formal group law to the multiplicative one over MU^*, and Theorem D states that the only one is 0. The same statement explains why KU *does* admit the twist: over KU^* the formal group law is already multiplicative, the identity homomorphism is available, and it realizes the Atiyah-Segal twist.

To see this concretely, a twist $f\colon K(\mathbb{Z},3) \to B\mathrm{GL}_1(E)$ loops to an H-map $\Omega f\colon \mathbb{CP}^\infty \to \mathrm{GL}_1(E)$, that is to an assignment of a unit $u(L) \in E^0(X)^\times$ to each line bundle L with $u(L\otimes L') = u(L)u(L')$, or said differently an action of $\mathrm{Pic}\,(X)$ on the theory by units. K-theory carries such an action, $L \mapsto [L]$, because a line bundle is an invertible object of the category KU is built from. In formal-group terms, the law of KU is multiplicative, $F(u,v) = u+v+uv$, which for $c_1^{KU}(L) = [L]-1$ is the identity $[L\otimes L'] = [L][L']$. Complex cobordism instead attaches to L the class $c_1^{MU}(L) \in MU^2(X)$, which is not a unit of $MU^0(X)$, and converts $\otimes$ into the *universal* law rather than into multiplication. Hence the objects MU is built from (proper maps with complex-oriented normal data) carry no tensor action of $\mathrm{Pic}\,(X)$. Proposition 4.2 is the precise form of this: such an H-map is a homomorphism from the universal formal group law to the multiplicative one over MU^*, and the only one is zero.

For the obstruction itself, because Ωf is null the twist cannot move the primary cobordism differential, hence cannot deform the primary MU-obstruction, that is the very thing a twisted MU would have had to deform:

Theorem 4.3: Twist-Invariance of MU

Let $[H] \in H^3(X;\mathbb{Z})$ and let $\tau = f\circ h\colon X \to B\mathrm{GL}_1(MU)$ be any twist pulled back from $K(\mathbb{Z},3)$, so h classifies $[H]$ and $f\colon K(\mathbb{Z},3) \to B\mathrm{GL}_1(MU)$ is arbitrary. Then the twisted MU-Atiyah-Hirzebruch differential is undeformed,

$$d_3^{MU,\tau} = \mathrm{Sq}^3_{\mathbb{Z}} = d_3^{MU}$$

Consequently the *primary* (d_3-level) twisted complex-cobordism obstruction coincides with the untwisted one for every $[H]$, and whenever $2i+5 > \dim_{\mathbb{R}} X$, the *full* twisted obstruction on $H^{2i}(X;\mathbb{Z})$, the conjunction of all differentials out of $E_r^{2i,0}$, coincides with the untwisted one.

Proof **:**
Let E be a ring spectrum with $\pi_{\mathrm{odd}}E = 0$, as holds for KU and MU, and let $\tau = f\circ h$ be a twist of E pulled back from $K(\mathbb{Z},3)$. The twist factors through the 2-connected $K(\mathbb{Z},3)$, so it acts trivially on the coefficients and the twisted spectral sequence has the same E_2-page as the untwisted one, with $E_2 = E_3$ since the odd rows vanish. For the construction and the formal properties of that spectral sequence (linearity over π_*E, naturality in (X,τ), normalisation at the zero twist) see [17, Thm. 18, Prop. 19], stated there for the differential refinement of which this is the underlying topological case, and [8, section 4] for $E = KU$. Only the differentials can move.

The difference $d_3^\tau - d_3$ is additive in the input x and natural in the pair (x,η), where $\eta \in H^3(-;\mathbb{Z})$ classifies the twist. Representing x and η by maps to Eilenberg-MacLane spaces, such an operation is an element of $H^{p+3}(K(\mathbb{Z},p)\times K(\mathbb{Z},3);\pi_2E)$, and it vanishes when $\eta=0$ (normalisation) and when $x=0$ (additivity). Since $H^4(K(\mathbb{Z},3);\pi_2E)$ and $H^5(K(\mathbb{Z},3);\pi_2E)$ vanish, the Künneth decomposition leaves only the term linear in both variables,

$$d_3^\tau - d_3 = \lambda\,(\eta\smile -),\qquad \lambda\in\pi_2E.$$

This is the universal-operations argument of Atiyah and Segal [8, (4.1)-(4.2)], carried out there for $E=KU$. To pin down λ, evaluate on $X=S^3$: the twisted spectrum is then clutched by $\tau|_{S^2}$, the spectral sequence reduces to the long exact sequence of the pair (S^3,pt), and d_3 is the boundary map of that clutching, so λ is the image of the generator of $\pi_3K(\mathbb{Z},3)$ under $\pi_3(f)$ (this is the computation of [8, Prop. 4.6]). Hence

$$d_3^\tau = d_3 + w_3(\tau)\smile(-),\qquad w_3(\tau)=h^*(\kappa_f)\in H^3(X;\pi_2E),$$

where $\kappa_f\in H^3(K(\mathbb{Z},3);\pi_2E)=\mathrm{Hom}(\pi_3K(\mathbb{Z},3),\pi_2E)$ is the class corresponding to $\pi_3(f)$, that is, the pullback along f of the fundamental class of the bottom Postnikov layer of $B\mathrm{GL}_1E$, with the coefficient identifications $\pi_3B\mathrm{GL}_1E\cong\pi_2\mathrm{gl}_1E\cong\pi_2E$, which is $\mathbb{Z}$ for both KU and MU (in topological grading, $\pi_2E=E^{-2}(\mathrm{pt})$).

The two theories differ only in λ. For KU the Atiyah-Segal twist is an isomorphism on π_3, so λ is a generator of $\pi_2KU=\mathbb{Z}$, $w_3([H])=\pm[H]$, and $d_3^{[H]}=\mathrm{Sq}^3_{\mathbb{Z}}+[H]\smile(-)$ up to the sign discussed in section 2 [8, Prop. 4.6]. For MU, proposition 4.2 gives $\Omega f\simeq *$ for every f, so $\pi_3(f)=\pi_2(\Omega f)=0$, hence $w_3(\tau)=0$ and

$$d_3^{MU,\tau}=\mathrm{Sq}^3_{\mathbb{Z}}\colon H^{2i}(X;\mathbb{Z})\longrightarrow H^{2i+3}(X;\pi_2MU)\cong H^{2i+3}(X;\mathbb{Z})$$

for every $[H]$, with no torsion hypothesis. The untwisted anchor $d_3^{MU}=\mathrm{Sq}^3_{\mathbb{Z}}$ is the same statement at $\tau=0$: the map $MU\to ku$ is an isomorphism on π_0 and π_2, so the two spectra share the first k-invariant identified in section 1.

For the last claim, a higher differential d_r with $r\geq 5$ out of $E_r^{2i,0}$ lands in a subquotient of $H^{2i+r}(X;\pi_{r-1}MU)$. When $2i+5>\dim_{\mathbb{R}}X$ this group vanishes for every $r\geq5$, so no higher differential acts, on either the twisted or the untwisted side, and the two full obstructions agree.

Note how this result is specifically about d_3 and not unconditionally on all d_r. Proposition 4.2 asserts that Ωf is null-homotopic, *not* that f is null. A ghost twist may carry non-trivial Postnikov invariants in the higher torsion groups $H^{2k+1}(K(\mathbb{Z},3);\pi_{2k}MU)$, which could in principle affect the differentials d_r with $r\geq5$. Vanishing of the primary correction therefore does not automatically force the higher differentials to remain undeformed. Indeed, for twisted KU with real coefficients the higher twisted differentials are iterated Massey products, $d_5(x)=-\{\eta,\eta,x\}$, $d_7(x)=-\{\eta,\eta,\eta,x\}$, … [8, Prop. 7.5], secondary and higher operations in the twist rather than functions of the primary class. Consequently theorem 4.3 establishes the deformation-invariance unconditionally at the d_3-level, and the invariance of the full obstruction is asserted only in the range where higher differentials vanish for degree reasons.

What is used below is the d_3-level statement. For every $[H]$ the primary differential of any twisted MU is $\mathrm{Sq}^3_{\mathbb{Z}}$, so the primary complex-cobordism obstruction is *twist-invariant*. The bounded-degree clause is not invoked below. The example of section 5 is a classifying space, of infinite real dimension, so $2i+5>\dim_{\mathbb{R}}X$ holds for no i and only the primary statement applies there.

Remark 4.4 (Where the dimension bound applies). Let X be a smooth projective threefold, so $\dim_{\mathbb{R}}X=6$ and $2i+5\geq7>6$ for every $i\geq1$. Then theorem 4.3 holds in its full form on X: for every $[H]\in H^3(X;\mathbb{Z})$ and every twist τ the twisted complex-cobordism obstruction on $H^{2i}(X;\mathbb{Z})$ is the untwisted one, and since no d_r with $r \geq 5$ can act,

$$d_3^{MU,\tau} = \mathrm{Sq}^3_{\mathbb{Z}}$$

is the *entire* cobordism obstruction. On a smooth projective fourfold, where $\dim_{\mathbb{R}} X = 8$, the same holds in codimension $i \geq 2$. Such an X carrying a nonzero twist exists already among threefolds: the Artin-Mumford double solid[13] has nonzero 2-torsion in $H^3(X;\mathbb{Z})$.

4.2 The Twisted K-theory Obstruction

Unlike twisted cobordism, twisted K-theory does acquire a deformed differential: by section 2 the deformation is $[H] \smile (-)$, which is nonzero as an operation whenever $[H] \neq 0$, since it is nonzero on $1 \in H^0(X;\mathbb{Z})$. This section shows what the deformation can and cannot detect: on the algebraic classes the operation either fails to vanish, in which case it is no obstruction, or it agrees there with the untwisted differential.

Definition 4.5: The Twisted Obstruction

Fix a twist $[H] \in H^3(X;\mathbb{Z})_{\mathrm{tors}}$. The *twisted obstruction* in codimension i is the first differential of the twisted spectral sequence, read as an operation on integral cohomology,

$$o_{[H]} \colon H^{2i}(X;\mathbb{Z}) \longrightarrow H^{2i+3}(X;\mathbb{Z}), \qquad o_{[H]}(\beta) = d_3^{[H]}\beta = \mathrm{Sq}^3_{\mathbb{Z}}\beta + [H] \smile \beta$$

A class β lifts to the twisted theory $KU_{[H]}(X)$ only if it is a permanent cycle of the twisted spectral sequence, and the first condition for this is $o_{[H]}(\beta) = 0$. An obstruction is only an obstruction if the algebraic classes satisfy it, and on those the Steenrod term drops out.

Proposition 4.6: The Weak-Lift Detection Criterion

Fix $i \geq 0$ and $[H] \in H^3(X;\mathbb{Z})$, and suppose there is an additive homomorphism $\psi_X \colon \mathrm{CH}^i(X) \to F^{2i}KU^0_{[H]}(X)$ whose classes in $E_\infty^{2i,0}$ are the classes of cl_i. Then

$$[H] \smile \mathrm{im}(\mathrm{cl}_i) = 0$$

Proof **:**
The class $\psi_X(z)$ determines an element of $E_\infty^{2i,0}$, a subquotient of $E_2^{2i,0} = H^{2i}(X;\mathbb{Z})$, and by hypothesis that element is the class of $\mathrm{cl}_i(z)$. For $\mathrm{cl}_i(z)$ to pass to this subquotient it must be a permanent cycle of the twisted spectral sequence. In particular, the first differential vanishes on it,

$$d_3^{[H]}\,\mathrm{cl}_i(z) = 0$$

The Atiyah-Hirzebruch obstruction gives $\mathrm{Sq}^3_{\mathbb{Z}}\,\mathrm{cl}_i(z) = 0$ (section 1), so

$$0 = d_3^{[H]}\,\mathrm{cl}_i(z) = \mathrm{Sq}^3_{\mathbb{Z}}\,\mathrm{cl}_i(z) + [H] \smile \mathrm{cl}_i(z) = [H] \smile \mathrm{cl}_i(z)$$

for every $z \in \mathrm{CH}^i(X)$, as we sought to show.

[13] The double cover of $\mathbb{P}^3$ branched along a quartic symmetroid $\det(\sum_i \lambda_i q_i) = 0$ attached to a base-point-free web of quadrics, with its ten nodes blown up. Artin and Mumford compute that $H^3(X;\mathbb{Z})$ contains an element of order two and use its birational invariance to exhibit a unirational, non-rational threefold [6].

Taken over all codimensions at once the criterion collapses, because the algebraic classes include the unit:

Corollary 4.7: The Unit Fails the Criterion

For every nonzero twist $[H]$ the fundamental class violates the criterion in codimension 0: with $1 = \mathrm{cl}_0([X]) \in H^0(X;\mathbb{Z})$,

$$o_{[H]}(1) = \mathrm{Sq}^3_{\mathbb{Z}}(1) + [H] \smile 1 = [H] \neq 0$$

Hence $[H] \smile \mathrm{im}(\mathrm{cl}) = 0$ across all codimensions holds only for $[H] = 0$.

Proof **:**
The Steenrod operations annihilate the unit, so $\mathrm{Sq}^3_{\mathbb{Z}}(1) = 0$ and $[H] \smile 1 = [H]$. As $1 = \mathrm{cl}_0([X])$ is algebraic, proposition 4.6 in codimension 0 forces $[H] = 0$, completing the proof.

Remark 4.8 (The failure is quantifiable). Corollary 4.7 is the $m = 1$ case of a divisibility computed in proposition 7.6: the image of the degree-zero edge homomorphism of the twisted spectral sequence is $\mathrm{ind}(\alpha) \cdot \mathbb{Z} \subseteq \mathrm{per}(\alpha) \cdot \mathbb{Z}$, so the rank-one class passes only when the Brauer class has period one. There the same computation is a theorem about the twisted conjecture rather than a failure of the untwisted one.

Codimension 0 is where the integral Hodge conjecture is trivially true, so the nontrivial test fixes a positive codimension i:

$$S_i = \left\{ [H] \in H^3(X;\mathbb{Z})_{\mathrm{tors}} \;:\; [H] \smile \beta = 0 \;\text{ for all } \beta \in \mathrm{im}(\mathrm{cl}_i) \right\}$$

a subgroup of $H^3(X;\mathbb{Z})_{\mathrm{tors}}$ that can be nonzero and that depends on X through the algebraic subgroup $\mathrm{im}(\mathrm{cl}_i)$. Call $[H]$ *admissible* in codimension i when $[H] \in S_i$. Once $\mathrm{im}(\mathrm{cl}_i)$ is known, admissibility is a cohomological condition on $[H]$, such as a spectral-sequence degeneration that kills the cup product against $\mathrm{im}(\mathrm{cl}_i)$. No description of S_i independent of the algebraic classes is available, since $\mathrm{im}(\mathrm{cl}_i)$ is *itself* the unknown of the integral Hodge conjecture in codimension i. How tightly the two can be bound is shown in the example of section 5, where $[H] \smile c_3 \equiv 0 \pmod{[H]^3}$ for spectral-sequence reasons (proposition 5.3) yet admissibility of the gerbe class on $\mathbb{Z}c_3$ is either empty or equivalent to the non-algebraicity of c_3.

Even so, on the algebraic classes an admissible twist coincides with the untwisted obstruction. For $[H] \in S_i$ the term $[H] \smile (-)$ vanishes on $\mathrm{im}(\mathrm{cl}_i)$, so $o_{[H]}$ and $\mathrm{Sq}^3_{\mathbb{Z}}$ agree there, and the only classes the twist could newly detect as non-algebraic are Hodge classes outside $\mathrm{im}(\mathrm{cl}_i)$ on which $[H] \smile (-)$ is nonzero.

4.3 Admissibility on a Single Class

The condition $[H] \in S_i$ ranges over all algebraic classes at once. Restricting it to a single algebraic class exposes what a twist can detect: on the line $\mathbb{Z}\alpha$ spanned by one class, admissibility is a single congruence, and that congruence reduces to an algebraic-divisibility question about α.

Say $[H]$ is *admissible on* a subgroup $S \subseteq H^{2i}(X;\mathbb{Z})$ if $o_{[H]}$ annihilates $\mathrm{im}(\mathrm{cl}_i) \cap S$. The $S = H^{2i}$ case is admissibility itself.

Proposition 4.9: A Torsion Twist Detects Only Modulo Its Order

Let X be smooth, $[H] \in H^3(X;\mathbb{Z})$ a twist of order ℓ, and $\alpha \in H^{2i}(X;\mathbb{Z})$. Then

$$o_{[H]}(m\alpha) = (m \bmod N)\, o_{[H]}(\alpha)$$

where $N = \operatorname{ord} o_{[H]}(\alpha)$ divides $\operatorname{lcm}(2,\ell)$.

Thus, whether $[H]$ is admissible on $\mathbb{Z}\alpha$ is decided by a single congruence modulo N on the algebraic multiples of α, and a torsion twist can detect no divisibility of those multiples by an integer prime to N.

Proof **:**
Both terms of $o_{[H]}(\alpha) = \mathrm{Sq}^3_{\mathbb{Z}}\alpha + [H] \smile \alpha$ are torsion: $\mathrm{Sq}^3_{\mathbb{Z}}\alpha$ lies in the image of the integral Bockstein, so is 2-torsion, and $[H] \smile \alpha$ is killed by ℓ since $\ell[H] = 0$. Their sum has order $N \big| \operatorname{lcm}(2,\ell)$, and additivity gives $o_{[H]}(m\alpha) = m\, o_{[H]}(\alpha) = (m \bmod N)\, o_{[H]}(\alpha)$. Admissibility on $\mathbb{Z}\alpha$ is the vanishing of this on $\operatorname{im}(\mathrm{cl}_i) \cap \mathbb{Z}\alpha$, which cannot decide divisibility by any integer prime to N, completing the proof.

Proposition 4.9 localizes the twist to a single modulus N. The admissibility question on $\mathbb{Z}\alpha$ then reduces to how the algebraic multiples of α sit against that modulus:

Proposition 4.10: Twisted Admissibility on $\mathbb{Z}\alpha$ Is an Algebraic Divisibility

Let X be smooth, $[H] \in H^3(X;\mathbb{Z})$ a twist of order ℓ, and $\alpha \in H^{2i}(X;\mathbb{Z})$. Let $N = \operatorname{ord} o_{[H]}(\alpha)$, and let d be the least positive integer with $d\alpha$ an algebraic class, or $d = \infty$ if no positive multiple of α is algebraic. Then $[H]$ is admissible on $\mathbb{Z}\alpha$ if and only if

$$N \big| d,$$

where $\operatorname{ord} 0 = 1$ (the zero class has additive order one) and $N \big| \infty$ is taken to hold when $d = \infty$.

Proof **:**
The operation $o_{[H]} = \mathrm{Sq}^3_{\mathbb{Z}} + [H] \smile (-)$ is additive, so $o_{[H]}(m\alpha) = m\, o_{[H]}(\alpha)$, and since $N = \operatorname{ord} o_{[H]}(\alpha)$ this vanishes if and only if $N \big| m$ (with $N = 1$ when $o_{[H]}(\alpha) = 0$). The twist is admissible on $\mathbb{Z}\alpha$ exactly when $o_{[H]}$ annihilates $\operatorname{im}(\mathrm{cl}_i) \cap \mathbb{Z}\alpha$, that is when $N \big| m$ for every m with $m\alpha$ an algebraic class. These m form a subgroup of $\mathbb{Z}$: it is $d\mathbb{Z}$ when some positive multiple of α is algebraic, in which case the condition is $N \big| d$, and it is $\{0\}$ otherwise, in which case the condition is vacuous, matching $N \big| \infty$. This proves the criterion.

When $o_{[H]}(\alpha) = 0$ the criterion holds with $o_{[H]} \equiv 0$ on $\mathbb{Z}\alpha$, that is, admissible but empty. On the algebraic multiples the Steenrod term drops by Atiyah-Hirzebruch, so there

$$o_{[H]}(m\alpha) = m\, [H] \smile \alpha$$

and N may equally be taken as $\operatorname{ord}([H] \smile \alpha)$, which divides ℓ. The divisibility d is the algebro-geometric data recording which multiples of α are algebraic, and determining it can be very difficult, as Totaro's analysis of classifying spaces shows [2].

5 Example: Twist-Invariance on $B(\mathrm{SL}_6/\mu_2)$

This section evaluates the twisted obstruction on a classifying space to which Totaro's smooth projective approximations and the factorization (1) apply [2], and whose codimension-three cycle map is the open instance of the integral Hodge conjecture studied by Tripathy [44]. The group is chosen minimally for the question: the centre μ_2 of SL_6 produces a nonzero gerbe class in $H^3(BG;\mathbb{Z})$, and the degree-six Chern class c_3 sits exactly where the primary differential $d_3\colon H^6 \to H^9$ acts. In Tripathy's analysis the classes obtained from representations of G supply only even multiples of c_3, while c_3 itself passes the cobordism obstruction, so its algebraicity is undecided by the untwisted invariants [44]. It is therefore the natural test case for whether the twisted differential decides anything new.

Throughout let $G = \mathrm{SL}_6/\mu_2$, let $\pi\colon BG \to K(\mathbb{Z}/2, 2)$ classify the gerbe, and let v be the generator of $H^3(K(\mathbb{Z}/2, 2);\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$, so that $[H] = \pi^*v$ is the gerbe class of the μ_2-extension, of order two in $H^3(BG;\mathbb{Z})$ (lemma 5.1). The letter x_1 is reserved for Gu's degree-three generator of $H^*(BPU_n;\mathbb{Z})$, used from proposition 5.4 on. Write W for the tautological six-dimensional representation of SL_6 and

$$c_3 = c_3(W) \;\in\; H^6(B\mathrm{SL}_6;\mathbb{Z})$$

its degree-six Chern class, viewed on BG through any lift $\tilde{c}_3 \in H^6(BG;\mathbb{Z})$ along the restriction $\iota^*\colon H^6(BG;\mathbb{Z}) \to H^6(B\mathrm{SL}_6;\mathbb{Z})$ to the fibre $\iota\colon B\mathrm{SL}_6 \hookrightarrow BG$ of π. It is the obstruction-bearing class of the example: W itself is not a representation of G, and the even exterior powers of W which are representable contribute only even multiples of c_3 (proposition 5.8), while c_3 passes the untwisted obstruction (proposition 5.4). Whether c_3 is algebraic on BG is the open codimension-three case of the integral Hodge conjecture there. Two computations give $o_{[H]}(c_3)$: a Serre spectral-sequence calculation and a comparison through BPU_6. Theorem 5.7 generalizes the mechanism to the family SL_n/μ_2, n even.

We first record the shape of the two cohomology groups involved.

Lemma 5.1: Low-degree cohomology of BG

Let $G = \mathrm{SL}_6/\mu_2$. Then

1. $H^3(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]\}$
2. $H^9(BG;\mathbb{Z})$ is 2-primary torsion
3. $F^1H^6(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]^2\}$, and $H^6(BG;\mathbb{Z})/F^1 \subseteq H^6(B\mathrm{SL}_6;\mathbb{Z}) = \mathbb{Z}c_3$
4. If moreover c_3 passes to $E_\infty^{0,6}$, as proposition 5.4 shows, then $H^6(BG;\mathbb{Z}) = \mathbb{Z}\{\tilde{c}_3\} \oplus \mathbb{Z}/2\mathbb{Z}\{[H]^2\}$ for any lift $\tilde{c}_3$ of c_3

Proof **:**

Work in the Serre spectral sequence of $B\mathrm{SL}_6 \to BG \xrightarrow{\pi} K(\mathbb{Z}/2, 2)$. The group $H^p(K(\mathbb{Z}/2, 2);\mathbb{Z})$ is a finite 2-group for $p > 0$, so every entry $E_2^{p,q}$ with $p > 0$ is one, while the column $E_2^{0,q} = H^q(B\mathrm{SL}_6;\mathbb{Z})$ is free and vanishes in odd degrees, since $H^*(B\mathrm{SL}_6;\mathbb{Z}) = \mathbb{Z}[c_2, \dots, c_6]$. In low degrees[a]

$$H^1 = H^2 = H^4 = 0, \quad H^3 = \mathbb{Z}/2\mathbb{Z}\{v\}, \quad H^5 = \mathbb{Z}/4\mathbb{Z}, \quad H^6 = \mathbb{Z}/2\mathbb{Z}\{v^2\} \qquad \text{on } K(\mathbb{Z}/2, 2)$$

1. In total degree 3 the only possible contributions are $E_\infty^{3,0}$ and $E_\infty^{0,3} = H^3(B\mathrm{SL}_6;\mathbb{Z}) = 0$, and the only differential that could reach $(3, 0)$ starts at $E_3^{0,2} = H^2(B\mathrm{SL}_6;\mathbb{Z}) = 0$. Hence $H^3(BG;\mathbb{Z}) = H^3(K(\mathbb{Z}/2, 2);\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$, generated by $\pi^*v = [H]$.

2. In total degree 9 every contributing entry has q even, hence p odd and positive, so all of them are 2-groups.

3. In total degree 6 the entries with $p > 0$ are $(p, q) \in \{(2, 4), (4, 2), (6, 0)\}$, and the first two vanish because $H^2(K(\mathbb{Z}/2, 2); \mathbb{Z}) = H^4(K(\mathbb{Z}/2, 2); \mathbb{Z}) = 0$. Differentials out of $(6, 0)$ land in negative q, and those into $(6, 0)$ start at $E_3^{3,2} = 0$ and $E_5^{1,4} = 0$, so $E_\infty^{6,0} = H^6(K(\mathbb{Z}/2, 2); \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{v^2\}$, giving $F^1H^6(BG; \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]^2\}$ and $H^6(BG; \mathbb{Z})/F^1 = E_\infty^{0,6} \subseteq H^6(B\mathrm{SL}_6; \mathbb{Z}) = \mathbb{Z}c_3$.

4. If $E_\infty^{0,6} = \mathbb{Z}c_3$ then the extension $0 \to \mathbb{Z}/2\mathbb{Z}\{[H]^2\} \to H^6(BG; \mathbb{Z}) \to \mathbb{Z}c_3 \to 0$ splits, $\mathbb{Z}$ being free, which is the stated decomposition.

[a] From $H^*(K(\mathbb{Z}/2, 2); \mathbb{Z}/2) = \mathbb{Z}/2[u_2, u_3, u_5, u_9, \ldots]$ with $u_3 = \mathrm{Sq}^1 u_2$, $u_5 = \mathrm{Sq}^2 u_3$ (Serre [43]), the Bockstein $d_1 = \mathrm{Sq}^1$ satisfies $d_1 u_2 = u_3$, $d_1 u_3 = 0$ and $d_1 u_5 = \mathrm{Sq}^1\mathrm{Sq}^2 u_3 = \mathrm{Sq}^3 u_3 = u_3^2$. The d_1-pairs (u_2, u_3) and (u_5, u_3^2) give the $\mathbb{Z}/2$'s in degrees 3 and 6, and the surviving class u_2^2 supports d_2 onto $u_2u_3 + u_5$, giving the $\mathbb{Z}/4$ in degree 5.

Lemma 5.2: The Steenrod Term Vanishes on the Fibre

On $B\mathrm{SL}_n$ the integral Steenrod operation vanishes on the Chern classes: $\mathrm{Sq}^3_{\mathbb{Z}} \equiv 0$, and in particular $\mathrm{Sq}^3_{\mathbb{Z}} c_3 = 0$.

Proof **:**
$\mathrm{Sq}^3_{\mathbb{Z}} = \delta\, \mathrm{Sq}^2 \rho_2$ factors through the integral Bockstein δ, whose image is 2-torsion. The ring $H^*(B\mathrm{SL}_n; \mathbb{Z}) = \mathbb{Z}[c_2, \ldots, c_n]$ is a polynomial ring, hence torsion-free, so it has no nonzero 2-torsion and $\mathrm{Sq}^3_{\mathbb{Z}} \equiv 0$, completing the proof.

The obstruction $o_{[H]}$ acts on $H^6(BG; \mathbb{Z})$, which, unlike the fibre, carries torsion (the class $[H]^2$), so lemma 5.2 gives only that $\mathrm{Sq}^3_{\mathbb{Z}} \tilde{c}_3$ restricts to 0 on the fibre for any lift $\tilde{c}_3$, hence lies in $F^1H^9(BG; \mathbb{Z}) = F^3H^9(BG; \mathbb{Z})$, the equality because $E_\infty^{1,8}$ and $E_\infty^{2,7}$ vanish ($H^1(K(\mathbb{Z}/2, 2); M) = 0$ for every coefficient group, and the fibre has no odd cohomology). The cup-product term is computed in the spectral sequence.

Proposition 5.3: The Cup Product Carries No Chern Component

In the Serre spectral sequence of $B\mathrm{SL}_6 \to BG \xrightarrow{\pi} K(\mathbb{Z}/2, 2)$ the differential on the fibre Chern classes is

$$d_3(c_i) = (7 - i)\, v\, c_{i-1}$$

Hence vc_3 is a d_3-boundary, $E_\infty^{3,6} = 0$, and for every lift $\tilde{c}_3$ of c_3 the product $[H] \smile \tilde{c}_3$ has vanishing filtration-3 component:

$$[H] \smile \tilde{c}_3 \in F^5H^9(BG; \mathbb{Z})$$

Proof **:**
The central extension $\mu_2 \to \mathrm{SL}_6 \to G$ deloops to the fibration $B\mathrm{SL}_6 \to BG \to K(\mathbb{Z}/2, 2)$, on which the group $B\mu_2 = K(\mathbb{Z}/2, 1)$ acts fibrewise: the central multiplication $\mu_2 \times \mathrm{SL}_6 \to \mathrm{SL}_6$ classifies the coaction $E \mapsto E \otimes \chi$ twisting the universal rank-six bundle by the order-two line bundle χ, whose Chern class $c_1(\chi) = w$ generates $H^2(B\mu_2; \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$ and transgresses to v. For such a fibration the differential d_3 on a fibre class is the coefficient of w in its coaction, multiplied by the transgression v; this is the computation by which Gu determines the analogous

differentials for BPU_n [19]. Expanding the total Chern class of the rank-six universal bundle after twisting by χ,

$$c(E \otimes \chi) = \sum_{j=0}^{6} c_j(E)\,(1+w)^{6-j}$$

the coefficient of w in degree $2i$ is $(6-(i-1))\,c_{i-1} = (7-i)\,c_{i-1}$, so the transgressive differential is $d_3(c_i) = (7-i)\,v\,c_{i-1}$ (the sign is immaterial, since v has order two). Thus

$$d_3(c_4) = 3\,vc_3 = vc_3$$

so $vc_3 \in E_3^{3,6}$ is a boundary and $E_\infty^{3,6} = 0$. The spectral sequence is multiplicative, $[H] = \pi^* v$ has filtration 3, and $\tilde{c}_3$ has symbol c_3, so $[H] \smile \tilde{c}_3$ has symbol $v \otimes c_3$ in $\mathrm{gr}^3 H^9(BG;\mathbb{Z})$, a quotient of $E_\infty^{3,6} = 0$. Hence $[H] \smile \tilde{c}_3 \in F^4 H^9$, and $F^4 H^9 = F^5 H^9$ because $E_\infty^{4,5} = 0$, the fibre having no odd cohomology, so $[H] \smile \tilde{c}_3 \in F^5 H^9(BG;\mathbb{Z})$, as we sought to show.

The spectral sequence leaves two points open: whether the free class c_3 passes to E_∞ (the even exterior powers of W produce only *even* multiples of c_3), and the value of $[H] \smile \tilde{c}_3$ within F^5. Both follow from a second computation, through the projective unitary quotient.

Proposition 5.4: Comparison with BPU_6

Let $f\colon BG \to BPU_6$ be induced by the further quotient $G = \mathrm{SL}_6/\mu_2 \to PU_6$ by $\mu_6/\mu_2 \cong \mu_3$. Then $f^* x_1 = [H]$. The fibre $B\mu_3$ of f carries only odd torsion, so f is an isomorphism on 2-local cohomology,[a] and

$$H^9(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]^3\}, \qquad [H]^3 \neq 0.$$

Moreover c_3 is a permanent cycle of the Serre spectral sequence of proposition 5.3, and for every lift $\tilde{c}_3$,

$$[H] \smile \tilde{c}_3 \in \{0, [H]^3\}.$$

[a] The 2-localization inverts every odd prime: $H^*(-;\mathbb{Z}_{(2)}) = H^*(-;\mathbb{Z}) \otimes \mathbb{Z}_{(2)}$, and a map is a 2-local isomorphism when it induces an isomorphism there. A map whose fibre has only odd torsion is one, since that torsion becomes invisible after inverting the odd primes.

The input is Gu's computation of $H^*(BPU_n;\mathbb{Z})$ in degrees ≤ 10 [19, Thm. 1.1]: the ring is $\mathbb{Z}[e_2, \ldots, e_{j_n}, x_1, y_{3,0}, y_{2,1}]/I_n$ with $|e_i| = 2i$, $|x_1| = 3$, $|y_{3,0}| = 8$, $|y_{2,1}| = 10$, and I_n containing the relations nx_1, $\varepsilon_2(n)x_1^2$ with $\varepsilon_2(n) = \gcd(n, 2)$, and

$$e_3\,x_1 = 0$$

the class $e_3 \in H^6(BPU_n;\mathbb{Z})$ being the generator of infinite order, unique up to sign and up to $e_3 \mapsto e_3 + x_1^2$. Its restriction to BU_6 is pinned by Gu's second theorem [19, Thm. 1.3]: for $k \leq 12$ the image of $H^k(BPU_n;\mathbb{Z}) \to H^k(BU_n;\mathbb{Z})$ is the subgroup of classes primitive for the coaction $E \mapsto E \otimes L$ of $K(\mathbb{Z}, 2) = BS^1$ by tensoring with line bundles. In degree six and rank six that subgroup is $\mathbb{Z} \cdot (5c_1^3 - 18c_1c_2 + 27c_3)$,[14] so, as x_1 restricts to 0 on BU_6,

$$e_3\big|_{BU_6} = \pm(5c_1^3 - 18c_1c_2 + 27c_3), \qquad e_3\big|_{B\mathrm{SL}_6} = \pm\,27\,c_3$$

an odd multiple of c_3, since c_1 restricts to 0 on $B\mathrm{SL}_6$.

[14] With $c_i \mapsto \sum_j \binom{6-j}{i-j} c_j w^{i-j}$ the coefficients of w, w^2, w^3 in $5c_1^3 - 18c_1c_2 + 27c_3$ are $90c_1^2 - 18(5c_1^2 + 6c_2) + 108c_2 = 0$, $540c_1 - 810c_1 + 270c_1 = 0$ and $1080 - 1620 + 540 = 0$, so the class is primitive and $\gcd(5, 18, 27) = 1$, so it generates the rank-one lattice of primitives cut out by the vanishing of the coefficient of w.

***Proof* :**

The kernel of $G \to PU_6$ is $\mu_6/\mu_2 \cong \mu_3$, giving the fibration $B\mu_3 \to BG \xrightarrow{f} BPU_6$. The reduced cohomology of $B\mu_3$ is all 3-torsion, so the Serre spectral sequence of f with $\mathbb{Z}_{(2)}$-coefficients collapses onto its base row and f^* is an isomorphism on 2-local cohomology.

In particular f^* is injective on the 2-torsion of $H^3(BPU_6;\mathbb{Z}) = \mathbb{Z}/6\mathbb{Z}\{x_1\}$, so $f^*(3x_1) \neq 0$. Since $H^3(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]\}$ is killed by 2 (lemma 5.1), we get $2f^*x_1 = 0$, hence $f^*x_1 = f^*(3x_1) \neq 0$, and the only nonzero class is $[H]$: thus $f^*x_1 = [H]$.

In degree nine, Gu's presentation leaves 2-locally only the monomials x_1^3 and e_3x_1 (the generators $y_{3,0}$ and $y_{2,1}$, of degrees 8 and 10, contribute none), the latter zero by the relation, and $2x_1^2 = 0$ for even n, hence $H^9(BPU_6;\mathbb{Z})_{(2)} = \mathbb{Z}/2\mathbb{Z}\{x_1^3\}$. By lemma 5.1 the group $H^9(BG;\mathbb{Z})$ is 2-primary, so it equals its 2-localization and therefore $H^9(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]^3\}$, with $[H]^3 = f^*(x_1^3) \neq 0$.

Next, c_3 is a permanent cycle of the Serre spectral sequence of proposition 5.3. The restriction of f^*e_3 to the fibre is $(f \circ \iota)^*e_3 = e_3|_{B\mathrm{SL}_6} = \pm 27c_3$, so $\pm 27c_3$ lies in the image of ι^*, which consists of permanent cycles. Every differential on c_3 lands in a subquotient of $H^r(K(\mathbb{Z}/2,2);H^{7-r}(B\mathrm{SL}_6;\mathbb{Z}))$ with $r > 0$, a 2-primary group on which multiplication by 27 is invertible, so $d_r(27c_3) = 0$ forces $d_r(c_3) = 0$, inductively over the pages. Lifts $\tilde{c}_3$ therefore exist, and lemma 5.1(4) applies.

Fix a lift $\tilde{c}_3 \in H^6(BG;\mathbb{Z})$ of c_3. Then $\iota^*(f^*e_3) = \pm 27c_3 = \iota^*(\pm 27\tilde{c}_3)$ and $\ker \iota^* = F^1H^6(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]^2\}$, so $f^*e_3 = \pm 27\,\tilde{c}_3 + t$ with $t \in \mathbb{Z}/2\mathbb{Z}\{[H]^2\}$. Pulling back Gu's relation,

$$0 = f^*(e_3\,x_1) = \pm 27\,[H] \smile \tilde{c}_3 + [H] \smile t$$

and $[H] \smile t \in \{0,[H]^3\}$, so $27\,[H] \smile \tilde{c}_3 \in \{0,[H]^3\}$. As 27 is odd and $H^9(BG;\mathbb{Z})$ is killed by 2, multiplication by 27 is the identity there and $[H] \smile \tilde{c}_3 \in \{0,[H]^3\}$, completing the proof.

Corollary 5.5: The Twist Is Vacuous on c_3

The twisted obstruction on c_3 carries no Chern-class information: both terms of $o_{[H]}(\tilde{c}_3) = \mathrm{Sq}^3_{\mathbb{Z}}\tilde{c}_3 + [H] \smile \tilde{c}_3$ lie in $H^9(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]^3\}$ (proposition 5.4), so

$$o_{[H]}(\tilde{c}_3) \in \{0,[H]^3\},$$

and $[H]^3 = \pi^*(v^3)$ restricts to 0 on the fibre, so it contains no Chern class.

Remark 5.6 (The ambiguity modulo $[H]^3$). The class c_3 has no canonical lift to $H^6(BG;\mathbb{Z})$: lifts differ by $F^1H^6(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]^2\}$, and $[H] \smile [H]^2 = [H]^3$, so $[H] \smile c_3$ is defined only modulo $[H]^3$. Nothing here depends on the value: $[H]^3 = \pi^*(v^3)$ restricts to 0 on the fibre because v does, carries no Chern class for the same reason, and is 2-torsion, which is all proposition 4.9 uses.

The same coaction runs for every even n, and for $n \equiv 2 \pmod 4$ the 2-local cross-check goes through as well, giving the following generalization.

Theorem 5.7: The Family SL_n/μ_2, n Even

Let $G_n = \mathrm{SL}_n/\mu_2$ with n even, let $[H] \in H^3(BG_n;\mathbb{Z})$ be the gerbe class, and let c_3 be the degree-six Chern class of the tautological representation (lemma 5.1 and its proof apply verbatim to every even n).

1. For every even n and every lift $\tilde{c}_3$,

$$[H] \smile \tilde{c}_3 \in F^5H^9(BG_n;\mathbb{Z}).$$

2. Suppose moreover that $n \equiv 2 \pmod 4$ and that the degree-six infinite-order generator of $H^*(BPU_n;\mathbb{Z})$ restricts on $B\mathrm{SL}_n$ to an odd multiple of c_3. Both hold for $n = 6$, with coefficient 27 [19, Thms. 1.1, 1.3]. Then c_3 is a permanent cycle, lifts $\tilde{c}_3$ exist, $[H] \smile \tilde{c}_3$ is pinned modulo $[H]^3$ as in proposition 5.4, and admissibility of $[H]$ on $\mathbb{Z}\tilde{c}_3$ reduces as in proposition 5.8: either $o_{[H]}$ vanishes on $\mathbb{Z}\tilde{c}_3$, or admissibility there is the parity condition $2\big|d$ of proposition 4.10 on the algebraic divisibility d of $\tilde{c}_3$.

Proof **:**
The coaction of proposition 5.3 runs unchanged for every even n, giving $d_3(c_i) = (n+1-i)\, v\, c_{i-1}$, thus $d_3(c_3) = (n-2)\, vc_2 = 0$ and $d_3(c_4) = (n-3)\, vc_3 = vc_3$ (as $n-3$ is odd), so vc_3 is a d_3-boundary and $[H] \smile \tilde{c}_3 \in F^4 = F^5$ exactly as in proposition 5.3. This proves (1), which used only that n is even; for $n \equiv 0 \pmod 4$ the statement is conditional on the existence of a lift, which part (2) supplies when its hypotheses hold. For $n \equiv 2 \pmod 4$ the kernel $\mu_{n/2}$ of $G_n \to PU_n$ has odd order, so $f\colon BG_n \to BPU_n$ is a 2-local isomorphism as in proposition 5.4. Under the added restriction hypothesis the argument of proposition 5.4 applies verbatim, with 27 replaced by that odd coefficient: it shows c_3 is permanent and pins $[H] \smile \tilde{c}_3$ modulo $[H]^3$, and proposition 4.10 with $N\big|2$ then reduces admissibility on $\mathbb{Z}\tilde{c}_3$ to the stated dichotomy. For $n \equiv 0 \pmod 4$ the kernel has even order and the cross-check is unavailable, so only the F^5-containment remains, completing the proof.

For $n = 6$ the parity condition can be identified outright, which is the $\alpha = \tilde{c}_3$ case of proposition 4.10.

Proposition 5.8: Twisted Admissibility on $\mathbb{Z}c_3$ and the Algebraicity of c_3

Let $[H]$ be the gerbe class on BG, $G = \mathrm{SL}_6/\mu_2$, and $\tilde{c}_3 \in H^6(BG;\mathbb{Z})$ a lift of c_3. If $o_{[H]}(\tilde{c}_3) = 0$, the twist is admissible on $\mathbb{Z}\tilde{c}_3$ but vacuously so, the obstruction $o_{[H]}$ being identically zero there. Otherwise, $[H]$ is admissible on $\mathbb{Z}\tilde{c}_3$ if and only if $\tilde{c}_3$ is non-algebraic.

Proof **:**
Apply proposition 4.10 with $\alpha = \tilde{c}_3$ and $\ell = 2$, so $N\big|2$, and take $o_{[H]}(\tilde{c}_3) \in \{0, [H]^3\}$ from corollary 5.5. If $o_{[H]}(\tilde{c}_3) = 0$ then $N = 1$, the twist is admissible on $\mathbb{Z}\tilde{c}_3$, and $o_{[H]}$ vanishes identically there, detecting nothing. Suppose $o_{[H]}(\tilde{c}_3) = [H]^3 \neq 0$, so $N = 2$ and admissibility is the condition $2\big|d$.

The exterior square $\Lambda^2 W$ is a representation of G, since μ_2 acts on it trivially, so $c_3(\Lambda^2 W) \in H^6(BG;\mathbb{Z})$ is algebraic on Totaro's approximations. Its restriction to the fibre is computed with Chern roots. Let $a_1, \ldots, a_n$ be the roots of W and p_k the power sums, on $B\mathrm{SL}_n$ where $c_1 = 0$, so $p_1 = 0$ and $p_3 = 3c_3$ by Newton's identities. The roots of $\Lambda^k W$ are the sums $\sum_{i\in S} a_i$ over

k-element subsets S, so

$$p_3(\Lambda^k W) = \sum_{|S|=k} \Big(\sum_{i\in S} a_i\Big)^3$$

and expanding the cube sorts the terms into the monomial types a_i^3, $a_i^2 a_j$ $(i \neq j)$, and $a_i a_j a_l$ $(i < j < l)$, occurring with coefficients 1, 3, 6 and total multiplicities $\binom{n-1}{k-1}$, $\binom{n-2}{k-2}$, $\binom{n-3}{k-3}$: each binomial counts the subsets S containing the one, two, or three indices of the monomial. With $\sum_{i\neq j} a_i^2 a_j = p_2 p_1 - p_3 = -p_3$ and $6\sum_{i<j<l} a_i a_j a_l = p_1^3 - 3p_1p_2 + 2p_3 = 2p_3$, this yields

$$p_3(\Lambda^k W) = \kappa(n,k)\, p_3, \qquad \kappa(n,k) := \binom{n-1}{k-1} - 3\binom{n-2}{k-2} + 2\binom{n-3}{k-3}$$

an integer depending only on n and k, which records how p_3 multiplies under Λ^k. As $c_1(\Lambda^k W) = \binom{n-1}{k-1} c_1 = 0$, Newton's identities give back $c_3(\Lambda^k W) = \frac{1}{3} p_3(\Lambda^k W) = \kappa(n,k)\, c_3$, with $\kappa(n,2) = n-4$. For $n = 6$ the restriction of $c_3(\Lambda^2 W)$ to the fibre is therefore $2c_3$, so $c_3(\Lambda^2 W)$ is a lift of $2c_3$. Two lifts of $2c_3$ differ by an element of $F^1H^6(BG;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}\{[H]^2\}$, so

$$4\,\tilde c_3 = 2\,c_3(\Lambda^2 W)$$

independently of the lift, and $4\tilde c_3$ is algebraic. Therefore $d \big| 4$, so d is odd only when $d = 1$, and $2\big| d$ holds if and only if $d \neq 1$, that is if and only if $\tilde c_3$ is non-algebraic.

The two lifts occupy opposite outcomes, since $o_{[H]}(\tilde c_3 + [H]^2) = o_{[H]}(\tilde c_3) + [H]^3$: indeed $\mathrm{Sq}^3_{\mathbb{Z}}[H]^2 = 0$, as $[H]^2$ is pulled back from $K(\mathbb{Z}/2, 2)$, where Sq^2 of a square is the square of Sq^1 and $\mathrm{Sq}^1\mathrm{Sq}^1 = 0$, while $[H] \smile [H]^2 = [H]^3$. In neither case does the twist decide more than the untwisted algebraicity of c_3, completing the proof.

Remark 5.9 (From $\mathbb{Z}c_3$ to S_3). Proposition 5.8 concerns admissibility on the line $\mathbb{Z}\tilde c_3$, not membership in $S_3 = \{[H] : [H] \smile \mathrm{im}(\mathrm{cl}_3) = 0\}$. By lemma 5.1 every class of $H^6(BG;\mathbb{Z})$ is $m\tilde c_3 + \epsilon[H]^2$ with $m \in \mathbb{Z}$ and $\epsilon \in \{0,1\}$, and

$$[H] \smile \big(m\tilde c_3 + \epsilon[H]^2\big) = (m\lambda + \epsilon)\,[H]^3, \qquad \text{where } [H] \smile \tilde c_3 = \lambda[H]^3,\ \lambda \in \{0,1\}$$

If $\lambda = 1$ and $\tilde c_3$ is algebraic then $[H] \notin S_3$, so membership in S_3 forces the non-algebraicity of $\tilde c_3$. The converse needs in addition that no algebraic class carries $m \not\equiv \epsilon \pmod 2$, a condition on $\mathrm{im}(\mathrm{cl}_3)$ that the line $\mathbb{Z}\tilde c_3$ does not see. If $\lambda = 0$ then $[H] \in S_3$ if and only if no algebraic class has $\epsilon = 1$, that is if and only if $\mathrm{im}(\mathrm{cl}_3) \subseteq \mathbb{Z}\tilde c_3$. In both cases the algebraicity of the torsion class $[H]^2$ is a separate question from that of c_3.

Remark 5.10 (The family beyond $n = 6$). The identification of the parity condition $2\big|d$ with non-algebraicity in proposition 5.8 needs a power of two annihilating $\tilde c_3$ modulo the algebraic classes. For general even n the exterior powers $\Lambda^k W$ with k even are representations of G_n and give, by the computation of that proof, $\kappa(n,k)\tilde c_3$ modulo F^1, hence a multiple $2\kappa(n,k)\tilde c_3$ that is algebraic. So the identification holds whenever $\gcd_k \kappa(n,k)$, over even k with $0 < k < n$, is a power of two. This is the case for every even n with $6 \le n \le 22$ by direct computation, the value being 2 for $n \equiv 2 \pmod 4$ and 2^v with $v \ge 2$ for $n \equiv 0 \pmod 4$. Without it, only the parity statement of theorem 5.7 is available.

On c_3 the twist is therefore vacuous (corollary 5.5), and even setting that computation aside its admissibility on $\mathbb{Z}c_3$ is either empty or equivalent to the non-algebraicity of c_3 (proposition 5.8), which is the very question the integral Hodge conjecture asks on this space. What obstructs c_3 is reached by no torsion twist, and by section 3 and section 4 by no H^3-twist and no cycle-receiving topological refinement. It remains to say where the obstruction does live.

6 The Residual Obstruction is Algebro-Geometric

Theorem 3.2 was proved by passing to the analytic side and invoking Quillen's theorem. That universality has an algebraic counterpart, carried by Levine-Morel algebraic cobordism Ω^* [33], the initial oriented theory on $\mathrm{Sm}_{\mathbb{C}}$. This section identifies the obstruction that passes everything proved so far and shows that it is an invariant of a comparison map between the two cobordism theories.

Section 3 through section 5 constrain the middle and right-hand objects $MU^{2i}(X) \otimes_{MU^*} \mathbb{Z}$ and $H^{2i}(X;\mathbb{Z})$ of the factorization (1): twisted K-theory receives no cycle and every complex-oriented theory factors through MU (section 3), no twist of MU deforms its primary differential (theorem 4.3), and the bare twisted differential is either inadmissible as an obstruction (section 4) or equivalent to the untwisted algebraicity question (proposition 5.8). All of these probe the codomain of $\widetilde{\mathrm{cl}}$ or the cohomology H^*, both invariants of the space $X(\mathbb{C})$. By theorem 3.2 every complex-oriented theory factors through MU, so $MU^{2i}(X) \otimes_{MU^*} \mathbb{Z}$ is already the finest multiplicative invariant of that kind. The map $\widetilde{\mathrm{cl}}$ itself, its image and its cokernel, entered none of these arguments. By the universality of Ω^* it is the reduction of a comparison between two cobordism theories, one algebraic and one topological [33, Thm. 1.2.6], [31, Introduction].

Let Ω^* be Levine-Morel algebraic cobordism on $\mathrm{Sm}_{\mathbb{C}}$: $\Omega^i(X)$ is generated by the classes $[f\colon Y \to X]$ of projective morphisms with Y smooth of pure codimension i, it is the *universal* (initial) oriented cohomology theory on $\mathrm{Sm}_{\mathbb{C}}$ [33, Thm. 1.2.6], and $\Omega^*(\mathrm{pt}) = \mathbb{L}^* = MU^*$ is the Lazard ring [33, Thm. 1.2.7].

Analytification makes $MU^{2*}(-(\mathbb{C}))$ with its Gysin pushforwards an oriented cohomology theory on $\mathrm{Sm}_{\mathbb{C}}$, so initiality of Ω^* produces a unique morphism of oriented cohomology theories [31, Introduction]

$$\rho\colon \Omega^i(X) \longrightarrow MU^{2i}(X(\mathbb{C})), \qquad [f\colon Y \to X] \longmapsto f_*(1)$$

In general ρ is neither injective nor surjective: algebraically distinct $[f], [f']$ can share an underlying complex-cobordism class, and a topological class need not equal $f_*(1)$ for any algebraic f. The subgroup $\mathrm{im}\rho$ is not an invariant of the space $X(\mathbb{C})$ in any evident way: its generators are the classes $f_*(1)$ of projective morphisms from smooth varieties, so membership in it is a question about X as a variety and not about the homotopy type of $X(\mathbb{C})$. This is what makes the remaining obstruction an algebro-geometric one.

Algebraic cobordism becomes the Chow group after base change along the augmentation $MU^* \to \mathbb{Z}$ killing $MU^{<0}$, in particular The Levine-Morel comparison theorem gives an *isomorphism*

$$\Omega^i(X) \otimes_{MU^*} \mathbb{Z} \xrightarrow{\sim} \mathrm{CH}^i(X), \qquad [f\colon Y \to X] \longmapsto f_*[Y]$$

of oriented cohomology theories [33, Thm. 1.2.19]. Passing from Ω to CH forgets only the cobordism coefficients. Totaro's $\widetilde{\mathrm{cl}}$ is ρ read after this reduction.

Proposition 6.1: The Algebro-Geometric Factorization

Let X be smooth quasi-projective over $\mathbb{C}$. Then Totaro's refined cycle map factors as

$$\mathrm{CH}^i(X) \xleftarrow{\sim} \Omega^i(X) \otimes_{MU^*} \mathbb{Z} \xrightarrow{\rho \otimes_{MU^*} \mathbb{Z}} MU^{2i}(X(\mathbb{C})) \otimes_{MU^*} \mathbb{Z}, \qquad \widetilde{\mathrm{cl}} = \rho \otimes_{MU^*} \mathbb{Z}$$

the left arrow being the Levine-Morel isomorphism and ρ the unique morphism of oriented cohomology theories out of the initial one. Consequently

$$\begin{aligned} \mathrm{coker}(\widetilde{\mathrm{cl}}) &= \mathrm{coker}\big(\rho \otimes_{MU^*} \mathbb{Z}\big) = \mathrm{coker}(\rho) \otimes_{MU^*} \mathbb{Z} \\ &= \mathrm{coker}\big(\Omega^i(X) \to MU^{2i}(X(\mathbb{C})) \otimes_{MU^*} \mathbb{Z}\big). \end{aligned}$$

Proof **:**
The factorization is the definition of ρ together with the Levine-Morel isomorphism above: both maps send $[f\colon Y \to X]$ to $f_*(1)$, in CH^i and in $MU^{2i} \otimes_{MU^*} \mathbb{Z}$ respectively, so the composite carries $f_*[Y]$ to $f_*(1)$, which is Totaro's definition of $\widetilde{\mathrm{cl}}$ through a resolution [2]. For the cokernels, $- \otimes_{MU^*} \mathbb{Z}$ is right exact, so applying it to $\Omega^i(X) \xrightarrow{\rho} MU^{2i}(X(\mathbb{C})) \to \mathrm{coker}(\rho) \to 0$ gives $\mathrm{coker}(\rho \otimes_{MU^*} \mathbb{Z}) = \mathrm{coker}(\rho) \otimes_{MU^*} \mathbb{Z}$, and the first equality is the factorization. For the last, the composite $\Omega^i(X) \to MU^{2i}(X(\mathbb{C})) \to MU^{2i}(X(\mathbb{C})) \otimes_{MU^*} \mathbb{Z}$ is MU^*-linear into a target annihilated by the augmentation ideal, hence kills that ideal and factors through $\Omega^i(X) \otimes_{MU^*} \mathbb{Z} = \mathrm{CH}^i(X)$ as $\widetilde{\mathrm{cl}}$. The two maps therefore have the same image and the same cokernel.

Proposition 6.1 isolates the residual *algebro-geometric obstruction*, and that group decides algebraicity once the topological obstruction has been passed.

Corollary 6.2: The Cokernel Decides Algebraicity

Let $\alpha \in \mathrm{Hdg}^{2i}(X;\mathbb{Z})$ lie in the image of θ, that is, let α pass the untwisted obstruction of section 1. Then α is algebraic if and only if the image of the coset $\theta^{-1}(\alpha)$ in $\mathrm{coker}(\widetilde{\mathrm{cl}})$ contains zero.

Proof **:**
If some lift $\beta \in \theta^{-1}(\alpha)$ equals $\widetilde{\mathrm{cl}}(z)$, then $\alpha = \theta(\widetilde{\mathrm{cl}}(z)) = \mathrm{cl}_i(z)$ is algebraic. Conversely if $\alpha = \mathrm{cl}_i(z)$ then $\widetilde{\mathrm{cl}}(z)$ is a lift of α lying in $\mathrm{im}(\widetilde{\mathrm{cl}})$. So α is algebraic exactly when $\theta^{-1}(\alpha)$ meets $\mathrm{im}(\widetilde{\mathrm{cl}})$, that is when its image in the cokernel contains zero, completing the proof.

Two lifts differ by an element of $\ker\theta$, so the coset $\theta^{-1}(\alpha)$ is an affine space under $\ker\theta$ (the filtration ambiguity of the Atiyah-Hirzebruch spectral sequence) and the invariant attached to α is the whole coset, that is α is non-algebraic exactly when its image avoids zero. A class failing the untwisted obstruction is non-algebraic already. Membership in $\mathrm{im}(\widetilde{\mathrm{cl}})$ is realizability by ρ, a condition on the algebraic side, and by theorem 3.2 every complex-oriented refinement of the cycle map factors through MU, the *target* of ρ, hence cannot detect which classes lie in $\mathrm{im}\rho$.

Remark 6.3 (The comparison in motivic terms). In Levine's original paper [31] referenced in this section, he uses motivic homotopy theory, and the follow-up paper [30] compares classical and motivic homotopy. For X smooth quasi-projective over a field of characteristic zero, Levine's comparison theorem identifies algebraic cobordism with one bigraded piece of the motivic Thom spectrum MGL [31]:

$$\Omega^i(X) \xrightarrow{\sim} MGL^{2i,i}(X)$$

(stated there in Borel-Moore form, $\Omega_n(X) \cong MGL'_{2n,n}(X)$), under which ρ becomes the map $MGL^{2i,i}(X) \to MU^{2i}(X(\mathbb{C}))$ induced by Betti realization $\mathcal{SH}(\mathbb{C}) \to \mathcal{SH}$, which carries MGL to MU (cf. [30]). Proposition 6.1 then reads: the residual obstruction is the failure of that realization map to be surjective, in the single bidegree $(2i, i)$, after reduction along $MU^* \to \mathbb{Z}$. It is a statement about that bidegree alone: nothing proved here constrains $MGL^{p,q}$ for other (p, q), and none of it asserts a failure of fullness for the realization functor.

On $B(\mathrm{SL}_6/\mu_2)$ the class c_3 passes the untwisted obstruction (proposition 5.4), so it lifts along θ, and by proposition 5.8 the most an admissible twist can do on $\mathbb{Z}c_3$ is restate whether c_3 is algebraic, that is, whether some lift of c_3 maps to zero in the cokernel of proposition 6.1, both sides computed on

Totaro's smooth projective approximations of the classifying space [2]. The rest of the analysis must be done on the algebraic side, with the tools defined there rather than on $X(\mathbb{C})$.[15]

6.1 The Hodge-Filtered Refinement Does Not Reach It

Every theory governed by theorem 3.2 is represented by a spectrum, so its value on X depends only on the homotopy type of $X(\mathbb{C})$. However, a *Hodge filtration* singles out the Hodge classes inside $H^{2i}(X;\mathbb{Z})$, and it is is invisible to every such theory. This leaves one further candidate for a refinement of (1): a theory built from *MU together with* the Hodge filtration, in the way Deligne cohomology is built from ordinary cohomology. Such a theory exists, due to Hopkins and Quick [23]. Its value depends on the complex structure of X and not only on the homotopy type of $X(\mathbb{C})$, so it is not given by theorem 3.2 and it carries strictly more information than MU. This subsection explains what the extra information is and why it does not reach the cokernel of proposition 6.1.

The model is Deligne cohomology (see [46, Ch. 12]). For smooth projective X it sits in a short exact sequence:

$$0 \longrightarrow J^{2i-1}(X) \longrightarrow H^{2i}_{D}(X;\mathbb{Z}(i)) \longrightarrow \mathrm{Hdg}^{2i}(X;\mathbb{Z}) \longrightarrow 0$$

$$J^{2i-1}(X) = \frac{H^{2i-1}(X;\mathbb{C})}{F^i H^{2i-1}(X;\mathbb{C}) + H^{2i-1}(X;\mathbb{Z})}$$

with $J^{2i-1}(X)$ the intermediate Jacobian (a complex torus). The Deligne cycle map $\mathrm{CH}^i(X) \to H^{2i}_D(X;\mathbb{Z}(i))$ lifts the ordinary one. Its image in the quotient $\mathrm{Hdg}^{2i}(X;\mathbb{Z})$ is $\mathrm{im}(\mathrm{cl}_i)$ (the image of the ordinary cycle map), while on the kernel of cl_i (the homologically trivial cycles) it is the Abel-Jacobi map into the torus. The refinement therefore decides nothing new about *which* Hodge classes are algebraic. What it adds is an invariant of the cycles that vanish in cohomology.

Hopkins and Quick carry out the same construction with MU in place of $H\mathbb{Z}$: for a rationally even spectrum E they glue the topological $E^{2i}(X)$ against the Hodge-filtered de Rham complex of X and obtain groups $E^{2i}_{D}(i)(X)$, written $MU^{2i}_{\mathrm{Hdg}}(X)$ here for $E = MU$. For compact Kähler X, in particular for the smooth projective X considered here, these sit in the exact sequence [23, Thm. 4.13]

$$0 \longrightarrow J^{2i-1}_{MU}(X) \longrightarrow MU^{2i}_{\mathrm{Hdg}}(X) \longrightarrow \mathrm{Hdg}^{2i}_{MU}(X) \longrightarrow 0$$

whose outer terms play the same roles as before: $\mathrm{Hdg}^{2i}_{MU}(X) \subseteq MU^{2i}(X)$ is the subgroup of classes whose image in cohomology with complexified MU-coefficients lies in the i-th step of the Hodge filtration, and $J^{2i-1}_{MU}(X)$ is a complex torus built from the odd-degree cohomology, the cobordism analogue of the intermediate Jacobian. Algebraic cobordism maps to the middle term, refining the realization ρ exactly as the Deligne cycle class refines cl_i [23, Thm. 7.10] (constructed there through the logarithmic variant $MU_{\log}$, which for projective X is the theory above).

The refinement fails to reach the residual obstruction for the same reason as in the model. By [23, Cor. 7.12] the composite

$$\Omega^i(X) \longrightarrow MU^{2i}_{\mathrm{Hdg}}(X) \longrightarrow \mathrm{Hdg}^{2i}_{MU}(X)$$

is the realization ρ (denoted ϕ_{MU} there), so in the quotient the image of the algebraic classes is the ordinary MU-image $\rho(\Omega^i)$ and nothing new appears. The new invariant lives in the subgroup $J^{2i-1}_{MU}(X)$, which a class reaches only by vanishing in the quotient. The extra content of MU_{Hdg} is

[15] Tools such as the Steenrod operations on Chow groups and their obstructions to algebraic classes [11, 9], the motivic Steenrod operations [45] acting through the comparison of remark 6.3, and the algebraic Morava K-theories [42].

thus an Abel-Jacobi invariant of homologically trivial classes, while the cokernel of proposition 6.1 is an invariant of classes with nonzero image in cohomology, so the two refinements measure disjoint failures. A class such as the c_3 of section 5 is not homologically trivial, so on it MU_{Hdg} and MU carry the same information, and the Hodge-filtered refinement decides nothing that MU does not.

7 The Integral Hodge Conjecture for Twisted α-Sheaves

This section records the twisted counterpart of definition 3.1 and theorem 3.2, in the framework of Hotchkiss and of Hotchkiss-Perry [24, 25] and of Perry [40], and shows how the definitions and results of section 3 transfer. The conclusion is that the twisted differential, ineffective for the untwisted case, is the obstruction in this setting, and that the unit which is missing in proposition 3.7 is replaced by the index of the Brauer class.

Fix a Brauer class α on X with gerbe class $[H] \in H^3(X;\mathbb{Z})_{\mathrm{tors}}$, and write $\mathrm{Coh}(X,\alpha)$ for the category of α-twisted coherent sheaves, that is, the modules over an Azumaya algebra of class α (definition 2.1). The untwisted conjecture concerns the chain

$$K_0(X) \longrightarrow KU^0(X) \longrightarrow H^{2i}(X;\mathbb{Z})$$

of section 1, written with $K_0(X)$ rather than $\mathrm{CH}^i(X)$ because the tested objects are sheaves. The classical assignment $Z \mapsto [\mathcal{O}_Z]$ has no twisted analogue since a twisted sheaf supported on Z carries the further data of an $\alpha|_Z$-twisted module on Z, so α-twisted sheaves are not built from algebraic cycles alone. The twisted counterpart is therefore formulated directly on twisted sheaves:

$$K_0(X,\alpha) \longrightarrow KU^0_{[H]}(X) \longrightarrow H^{2i}(X;\mathbb{Z}), \qquad K_0(X,\alpha) = K_0\big(\mathrm{Coh}(X,\alpha)\big)$$

with the right-hand map the edge homomorphism $\theta_{[H]}$ of the twisted spectral sequence of section 2. The question is which classes come from $K_0(X,\alpha)$ through the twisted edge map, and the first condition for a class β to lie in its image is $d_3^{[H]}\beta = 0$. Here the twisted differential is the correct obstruction because the objects being lifted are themselves twisted. Hotchkiss and Perry compute the twisted topological K-theory $KU^0_{[H]}(X)$ from exactly this spectral sequence, and its graded pieces on a threefold [25, Lems. V.1, V.4], and Perry proves the corresponding integral Hodge statement for two-dimensional Calabi-Yau categories [40][16]. Neither conjecture implies the other: an untwisted cycle does not define a class in $KU^0_{[H]}(X)$, and an α-twisted sheaf has no canonical class in $KU^0(X)$, the two directions of the failure recorded in proposition 3.7. The definitions and results of section 3 nevertheless transfer, as the rest of this section shows.

7.1 The Twisted Cycle-Receiving Structure

The axioms of definition 3.1 transfer once the effects of a twist are encoded. First, proposition 3.7 locates the failure in the absence of anything canonical to push forward, not in the absence of a pushforward: the Gysin map $\iota_!$ of [13] exists, and $\iota_!(\mathcal{F})$ is defined for every $\iota^*[H]$-twisted bundle $\mathcal{F}$ on Z while $\iota_!(1)$ is not. The twisted structure therefore receives pairs $(Z,\mathcal{F})$ with $\mathcal{F} \in \mathrm{Vect}_{[H]|_Z}(Z)$, which is the codimension-$\geq i$ step of the support filtration on $K_0(X,\alpha)$. Second, the target is graded: the mismatch of section 3.2, that $[Z]_E \smile [W]_E$ lies over $2[H]$ while $[Z\cap W]_E$ lies over $[H]$, is repaired by carrying all twists at once, with the tensor rule of section 2 as the grading. Third, the normalisation is by rank, which is what replaces the unit.

[16] A two-dimensional Calabi-Yau category is a category with the formal properties of the derived category of a K3 or abelian surface (Serre duality holds) without being the derived category of any variety. Formulating the conjecture on the category makes the twisted sheaves themselves the tested objects, which is the proper setting for the twisted differential, and where the strategy this paper rules out for untwisted cycles is correctly applied.

Definition 7.1: Twist-Graded Ring Spectrum

Let E be an E_∞-ring spectrum and $f\colon K(\mathbb{Z},3) \to \mathrm{BGL}_1(E)$ an H-map. For $[H] \in H^3(X;\mathbb{Z})$ classified by $h\colon X \to K(\mathbb{Z},3)$, write $E^*_{[H]}(X)$ for the τ-twisted cohomology of definition 4.1 with $\tau = f \circ h$. Since f is an H-map, the smash product over E of the bundles of invertible E-modules classified by $f \circ h$ and $f \circ h'$ is the bundle classified by $f \circ (h + h')$, so there are pairings

$$E^p_{[H]}(X) \otimes E^q_{[H']}(X) \longrightarrow E^{p+q}_{[H+H']}(X)$$

associative and commutative, whose case $[H'] = 0$ is the $E^*(X)$-module structure of section 2. Call the pair $E_\bullet = (E, f)$ a *twist-graded ring spectrum* and write $E_0 = E$ for its untwisted piece.

Definition 7.2: Twisted Cycle-Receiving Structure

Let $E_\bullet$ be a twist-graded ring spectrum. A *twisted cycle-receiving structure* on $E_\bullet$ assigns to every smooth quasi-projective $X/\mathbb{C}$, every $[H] \in H^3(X;\mathbb{Z})$, every smooth closed subvariety $Z \subseteq X$ of codimension i, and every twisted bundle $\mathcal{F} \in \mathrm{Vect}_{[H]|_Z}(Z)$ a class

$$[Z, \mathcal{F}]_E \ \in \ (E_{[H]})^{2i}_Z(X)$$

in the cohomology of X with supports in Z, such that

1. *(base change)* for every morphism $g\colon X' \to X$ of smooth varieties transverse to Z, with restriction $g'\colon g^{-1}Z \to Z$,

   $$g^*[Z, \mathcal{F}]_E = [\,g^{-1}Z,\ g'^*\mathcal{F}\,]_E \quad \text{in } (E_{g^*[H]})^{2i}_{g^{-1}Z}(X'),$$

   and $[\emptyset, \mathcal{F}]_E = 0$

2. *(graded multiplicativity)* whenever Z, W are smooth of codimensions i, j and meet transversally, and $\mathcal{F} \in \mathrm{Vect}_{[H]|_Z}(Z)$, $\mathcal{G} \in \mathrm{Vect}_{[H']|_W}(W)$,

   $$[Z, \mathcal{F}]_E \smile [W, \mathcal{G}]_E = \big[\,Z \cap W,\ \mathcal{F}|_{Z\cap W} \otimes \mathcal{G}|_{Z\cap W}\,\big]_E \quad \text{in } (E_{[H+H']})^{2i+2j}_{Z\cap W}(X)$$

3. *(normalisation)* $[X, \mathcal{O}_X]_E = 1$ in $(E_0)^0_X(X) = E^0(X)$, and $\mathcal{F} \mapsto [X, \mathcal{F}]_E$ carries direct sums to sums

4. *(comparison)* the image $\overline{[Z, \mathcal{F}]}_E \in (E_{[H]})^{2i}(X)$ lies in F^{2i}. The class $\mathrm{rk}(\mathcal{F}) \cdot \mathrm{cl}_i([Z])$, viewed in $E_2^{2i,0} = H^{2i}(X; E^0)$ along the coefficient map induced by the unit $\mathbb{Z} = \pi_0\mathbb{S} \to E^0$, is a permanent cycle of the $[H]$-twisted spectral sequence, and the class of $\overline{[Z, \mathcal{F}]}_E$ in $E_\infty^{2i,0}$ is its class there.

The definition quantifies over all ambient varieties, so axiom (3) applied to Z as its own ambient space makes $\mathcal{F} \mapsto [Z, \mathcal{F}]_E \in (E_{[H]|_Z})^0(Z)$ additive, and theorem 7.3(2) below then gives additivity of $[Z, \mathcal{F}]_E$ in every codimension. Restricting to $[H] = 0$ and $\mathcal{F} = \mathcal{O}_Z$ recovers definition 3.1: axiom (2) becomes $[Z]_E \smile [W]_E = [Z \cap W]_E$ since $\mathcal{O} \otimes \mathcal{O} = \mathcal{O}$, axiom (3) becomes $[X]_E = 1$, and axiom (4) becomes the old comparison since $\mathrm{rk}\mathcal{O} = 1$.

Theorem 7.3: Twisted Cycle-Receiving Theorem

Let $E_\bullet$ carry a twisted cycle-receiving structure. Then

1. E_0 is complex-oriented, by a class $x \in \widetilde{E}^2(\mathbb{CP}^\infty)$ with associated ring map $\varphi_x \colon MU \to E_0$.

2. For a smooth variety Z and $\mathcal{F} \in \mathrm{Vect}_{[H]}(Z)$, let

$$c_{[H]}(\mathcal{F}) \;:=\; [Z, \mathcal{F}]_E \;\in\; (E_{[H]})^0(Z)$$

denote the intrinsic class obtained by evaluating the structure on Z as its own ambient space in codimension zero ($X = Z$, $i = 0$). Then for every smooth closed embedding $\iota \colon Z \hookrightarrow X$ of codimension i, under the identification $(E_{[H]})^{2i}_Z(X) \cong \widetilde{(E_{[H]})}^{2i}(\mathrm{Th}\, N_{Z/X})$,

$$[Z, \mathcal{F}]_E = \mathrm{th}^x(N_{Z/X}) \smile c_{[H]}(\mathcal{F}), \qquad \overline{[Z, \mathcal{F}]}_E = \iota_*\big(c_{[H]}(\mathcal{F})\big),$$

where th^x is the Thom class of the orientation of (1), the pullback to the Thom space is suppressed, and ι_* is the Gysin map acting on the E_0-module $E_{[H]}$.

Consequently the structure is determined by the orientation of E_0 *together* with the natural transformation

$$\mathrm{Vect}_{[H]}(-) \to (E_{[H]})^0(-), \qquad \mathcal{F} \mapsto c_{[H]}(\mathcal{F})$$

which is additive, satisfies

$$c_{[H]}(\mathcal{F}) \smile c_{[H']}(\mathcal{G}) = c_{[H+H']}(\mathcal{F} \otimes \mathcal{G})$$

has $c_0(\mathcal{O}) = 1$, and maps $\mathcal{F}$ to its rank $\mathrm{rk}(\mathcal{F}) \in E_\infty^{0,0}$ under the degree-zero edge map.

The statement is theorem 3.2 with the fundamental class $\iota_*(1)$ replaced by $\iota_*(c_{[H]}(\mathcal{F}))$. Assertion (2) again says that the structure *is* the Gysin system of an orientation. The twist contributes only the choice of the object pushed forward, which over $[H] \neq 0$ has no distinguished value.

Proof **:**
For (1), restrict the structure to $[H] = 0$ and $\mathcal{F} = \mathcal{O}_Z$. As noted above the four axioms then read as definition 3.1 for the homotopy-commutative ring spectrum E_0, so theorem 3.2(1) applies verbatim: the hyperplane classes $x_n = \overline{[H_n, \mathcal{O}]}_E \in E^2(\mathbb{P}^n)$ are independent of the hyperplane, reduced, and compatible under transverse linear embeddings, and they assemble to $x \in \widetilde{E}^2(\mathbb{CP}^\infty)$ restricting to the unit on $\mathbb{P}^1$, hence to φ_x. The orientation th built there from the classes $[0_V, \mathcal{O}]_E$ of zero sections, and its identification with th^x, likewise follows.

For (2), fix a smooth closed embedding $\iota \colon Z \hookrightarrow X$ of codimension i and $\mathcal{F} \in \mathrm{Vect}_{[H]|_Z}(Z)$, and write $N = N_{Z/X}$ with projection $p \colon N \to Z$ and zero section $z \colon Z \hookrightarrow N$.

Excision and the tubular neighbourhood theorem identify $(E_{[H]})^{2i}_Z(X)$ with $(E_{p^*([H]|_Z)})^{2i}_{0_N}(N)$, and under this identification $[Z, \mathcal{F}]_E$ corresponds to $[0_N, \mathcal{F}]_E$: the proof of lemma 3.4 applies verbatim to the pair $(Z \times \mathbb{A}^1,\ \mathrm{pr}^*\mathcal{F})$ on the deformation space, the twisted axiom (1) supplying the two transverse restrictions to the fibres.

The total space N is a smooth closed subvariety of itself of codimension 0 carrying $p^*\mathcal{F} \in \mathrm{Vect}_{p^*([H]|_Z)}(N)$, and it meets 0_N transversally with $0_N \cap N = 0_N$ and $\mathcal{O}_{0_N} \otimes (p^*\mathcal{F})|_{0_N} = \mathcal{F}$. Axiom (2) therefore gives

$$[\,0_N, \mathcal{O}_{0_N}\,]_E \smile [\,N, p^*\mathcal{F}\,]_E = [\,0_N, \mathcal{F}\,]_E$$

The first factor is untwisted and equals $\mathrm{th}(N) = \mathrm{th}^x(N)$ by (1). The second lies in $(E_{p^*([H]|_Z)})^0(N)$. The zero section z is transverse to the codimension-zero subvariety N and satisfies $z^*p^*\mathcal{F} = \mathcal{F}$, so axiom (1) gives $z^*[N, p^*\mathcal{F}]_E = [Z, \mathcal{F}]_E = c_{[H]}(\mathcal{F})$, and p^* is inverse to z^* by homotopy invariance, so $[N, p^*\mathcal{F}]_E = p^*c_{[H]}(\mathcal{F})$. Hence

$$[Z, \mathcal{F}]_E = \mathrm{th}^x(N) \smile p^*c_{[H]}(\mathcal{F})$$

which is the displayed formula with the pullback made explicit. Since $E_{[H]}$ is a module over the oriented E_0 and $\mathrm{th}^x(N)$ restricts on each fibre to $\mathrm{th}^x(\mathcal{O})^{\smile i}$, by Mayer-Vietoris the cup product with $\mathrm{th}^x(N)$ is an isomorphism

$$(E_{[H]})^*(Z) \xrightarrow{\sim} \widetilde{(E_{[H]})}^{*+2i}(\mathrm{Th}\, N)$$

exactly as for E_0 in the proof of theorem 3.2. The Gysin map ι_* of the orientation acting on $E_{[H]}$ is by definition that isomorphism followed by the forget-support map $\widetilde{(E_{[H]})}^{2i}(\mathrm{Th}\, N) \cong (E_{[H]})^{2i}_Z(X) \to (E_{[H]})^{2i}(X)$, which yields

$$\overline{[Z, \mathcal{F}]}_E = \iota_*\big(c_{[H]}(\mathcal{F})\big).$$

The listed algebraic properties of c are axioms (2) and (3) evaluated on Z in codimension zero: additivity and $c_0(\mathcal{O}) = 1$ are axiom (3), and $c_{[H]}(\mathcal{F}) \smile c_{[H']}(\mathcal{G}) = c_{[H+H']}(\mathcal{F} \otimes \mathcal{G})$ is axiom (2) with $Z = W = X$. Furthermore, applying axiom (4) to $Z \subseteq Z$ ($i = 0$) gives that the degree-zero edge homomorphism ε_Z sends $c_{[H]}(\mathcal{F})$ to $\mathrm{rk}(\mathcal{F}) \in E^{0,0}_\infty(Z) \subseteq H^0(Z; E^0)$.

Conversely, given an orientation on E_0 and a transformation c satisfying these properties, the formula $[Z, \mathcal{F}]_E := \mathrm{th}^x(N_{Z/X}) \smile p^*c_{[H]}(\mathcal{F})$ satisfies axioms (1)-(3) by naturality and multiplicativity of Thom classes together with $N_{Z\cap W/X} = N_{Z/X}|_{Z\cap W} \oplus N_{W/X}|_{Z\cap W}$. Finally, it satisfies axiom (4). The pushforward ι_* raises the skeletal filtration by $2i$, since its values are supported near Z and so vanish on the $(2i-1)$-skeleton of a cell structure dual to Z, and on the associated graded it induces the Gysin map of ordinary cohomology, which sends 1_Z to $\mathrm{cl}_i([Z])$: the pushforward is the composite of the Pontryagin-Thom collapse with cup product by the Thom class, and on the E_2-page the Thom class acts through its symbol, the ordinary Thom class. The edge homomorphism is therefore compatible with the pushforward:

$$\varepsilon_X\big(\overline{[Z, \mathcal{F}]}_E\big) = \iota_*\big(\varepsilon_Z(c_{[H]}(\mathcal{F}))\big) = \iota_*\big(\mathrm{rk}(\mathcal{F}) \cdot 1_Z\big) = \mathrm{rk}(\mathcal{F}) \cdot \mathrm{cl}_i([Z])$$

completing the proof.

Corollary 7.4: Twisted K-theory Receives Twisted Cycles

Let $KU_\bullet$ be KU with the Atiyah-Segal twist. Then $KU_\bullet$ is twist-graded and carries a twisted cycle-receiving structure, namely

$$[Z, \mathcal{F}]_{KU} := \iota_!(\mathcal{F}) \in (KU_{[H]})^{2i}_Z(X), \qquad c_{[H]}(\mathcal{F}) = [\mathcal{F}] \in KU^0_{[H]}(Z)$$

with $\iota_!$ the twisted Gysin map of [13, Thm. 1.1].

Proof **:**
The Atiyah-Segal twist is classified by the delooping of the H-space map $K(\mathbb{Z}, 2) \simeq BU(1) \to \mathrm{GL}_1(KU)$ sending $L \mapsto [L]$. Its classifying map $f\colon K(\mathbb{Z}, 3) \to B\mathrm{GL}_1(KU)$ is therefore an

H-map, equipping $KU_\bullet$ with associative and commutative pairings

$$KU_{[H]} \wedge_{KU} KU_{[H']} \longrightarrow KU_{[H+H']}$$

whose induced product on cohomology agrees with the tensor product of twisted bundles, and thus is a twist-graded structure.

Next, assigning to an $[H]$-twisted vector bundle $\mathcal{F}$ its class $[\mathcal{F}] \in KU^0_{[H]}(Z)$ defines a natural transformation

$$c_{[H]} \colon \mathrm{Vect}_{[H]}(Z) \to KU^0_{[H]}(Z)$$

that satisfies:

- *Additivity:* $[\mathcal{F} \oplus \mathcal{F}'] = [\mathcal{F}] + [\mathcal{F}']$.
- *Multiplicativity:* $[\mathcal{F}] \smile [\mathcal{G}] = [\mathcal{F} \otimes \mathcal{G}] \in KU^0_{[H+H']}(Z)$ under the pairing above.
- *Normalisation:* $c_0(\mathcal{O}_Z) = [\mathcal{O}_Z] = 1 \in KU^0(Z)$.
- *Edge compatibility:* In the $[H]$-twisted Atiyah-Hirzebruch spectral sequence, the degree-zero edge homomorphism $\varepsilon_Z \colon KU^0_{[H]}(Z) \to H^0(Z;\mathbb{Z}) = E_2^{0,0}(Z)$ is given on connected components by the topological rank, sending $[\mathcal{F}] \mapsto \mathrm{rk}(\mathcal{F}) \cdot 1_Z$.

For the Gysin maps, the untwisted piece $E_0 = KU$ carries its standard complex orientation $x = [L] - 1 \in \widetilde{KU}^2(\mathbb{CP}^\infty)$ with associated Thom class th^{KU}. For any smooth closed embedding $\iota \colon Z \hookrightarrow X$ of codimension i, the normal bundle $N = N_{Z/X}$ is complex, hence canonically Spin^c with vanishing third integral Stiefel-Whitney class $W_3(N) = 0$. By theorem 7.3, the orientation th^{KU} together with $c_{[H]}$ uniquely determines a twisted cycle-receiving structure whose supported class is

$$[Z, \mathcal{F}]_{KU} \;=\; \mathrm{th}^{KU}(N_{Z/X}) \smile p^*[\mathcal{F}] \;\in\; (KU_{[H]})^{2i}_Z(X).$$

Under the Thom identification $(KU_{[H]})^{2i}_Z(X) \cong \widetilde{(KU_{[H]})}^{2i}(\mathrm{Th}\, N_{Z/X})$, this class is the Carey-Wang pushforward $\iota_!(\mathcal{F})$ of [13], completing the proof.

Remark 7.5 (The graded structure on cobordism carries no twisted information). Let $E_\bullet$ be twist-graded with $E_0 = MU$, so that $E_{[H]}$ is the twisted cobordism of definition 4.1 for some f. By theorem 4.3, $d_3^{MU,\tau} = \mathrm{Sq}^3_{\mathbb{Z}}$ for every f and every $[H]$, so axiom (4) of definition 7.2 requires

$$\mathrm{Sq}^3_{\mathbb{Z}}(\mathrm{rk}(\mathcal{F})\, \mathrm{cl}_i([Z])) = 0$$

that is the untwisted condition. The twisted content of a twisted cycle-receiving structure is therefore carried by the deformation of d_3, which is nonzero for KU and zero for MU. Note that this does not say that $MU_\tau \simeq MU$: proposition 4.2 gives that Ωf is null, not that f is.

7.2 The Period-Index Bound

By theorem 7.3 the twisted structure is determined by the orientation of E_0 together with the transformation $F \mapsto c_{[H]}(F)$, and over $[H] \neq 0$ that transformation has no distinguished argument: there is no canonical twisted bundle on Z to push forward. What replaces the unit is therefore not an element but the numerical invariant recording which twisted bundles exist, namely the set of their ranks.

The *period* $\mathrm{per}(\alpha)$ is the order of α, equal to the order of $[H]$ in $H^3(X;\mathbb{Z})_{\mathrm{tors}}$. Write $\mathrm{ind}(\alpha)$ for the greatest common divisor of the ranks of the α-twisted bundles, equivalently of the degrees of the

Azumaya algebras of class α (definition 2.1). Over a field this is the index of the Brauer class in the usual sense [16], and on a space it is the topological index of [5]. The degree-zero corner of the twisted spectral sequence computes both and compares them.

Proposition 7.6: The Degree-Zero Corner Is the Period-Index Bound

Let X be connected and as in the Conventions (a smooth projective complex variety or one of Totaro's approximations) and let α have gerbe class $[H] \in H^3(X;\mathbb{Z})_{\mathrm{tors}}$. Then the edge homomorphism of the twisted spectral sequence in degree zero is the rank, so

$$E_\infty^{0,0} = \mathrm{ind}(\alpha)\cdot\mathbb{Z} \ \subseteq\ \mathrm{per}(\alpha)\cdot\mathbb{Z} \qquad \text{in } H^0(X;\mathbb{Z}) = \mathbb{Z}$$

and therefore $\mathrm{per}(\alpha)\,\big|\,\mathrm{ind}(\alpha)$.

Proof **:**
For connected X the edge homomorphism in degree zero is restriction to a point, that is the rank. By the definition of $KU^0_{[H]}(X)$ as $K_0(\mathrm{Vect}_{[H]}(X))$ in section 2, every class is a difference of α-twisted bundles, so the image of the rank is the subgroup of $\mathbb{Z}$ generated by their ranks, namely $\mathrm{ind}(\alpha)\cdot\mathbb{Z}$, and that image is $E_\infty^{0,0}$. For the other inclusion, $\mathrm{Sq}^3_{\mathbb{Z}}$ annihilates $H^0(X;\mathbb{Z})$, so $d_3^{[H]}(m) = m\,[H]$ by definition 4.5, which vanishes exactly when $\mathrm{per}(\alpha)\big|m$, hence

$$E_4^{0,0} \subseteq \mathrm{per}(\alpha)\cdot\mathbb{Z}$$

No differential enters the column $p = 0$, since its source would sit in $p = -r < 0$, so $E_\infty^{0,0}$ is a subgroup of $E_4^{0,0}$ and satisfies the same inclusion. Combining, we get

$$\mathrm{ind}(\alpha)\mathbb{Z} \subseteq \mathrm{per}(\alpha)\mathbb{Z}$$

completing the proof.

Note that this is a known result: the divisibility itself is the topological period-index inequality of [5], and the identification of the degree-zero graded piece with $\mathrm{ind}(\alpha)\cdot\mathbb{Z}$ is [25, Lem. V.4], where the higher graded pieces on a threefold are computed as well. Corollary 4.7 is the $m = 1$ case: the rank-one class is the unit, and it passes the twisted differential only when $\mathrm{per}(\alpha) = 1$. The obstruction that made the twist useless for untwisted cycles reappears here as an invariant with content: the smallest rank in which the Brauer class admits twisted bundles.

Note that proposition 7.6 is axiom (4) of definition 7.2 in codimension zero. For $Z = X$ and F of rank r the axiom requires $r\cdot\mathrm{cl}_0([X]) = r$ to be a permanent cycle of the $[H]$-twisted spectral sequence, so $d_3^{[H]}(r) = r[H] = 0$ and $\mathrm{per}(\alpha)\big|r$. Ranging over F gives $\mathrm{per}(\alpha)\big|\,\mathrm{ind}(\alpha)$. Where definition 3.1(3) asserts that 1 lies in the image of the degree-zero edge map, definition 7.2(4) asserts only that $\mathrm{ind}(\alpha)$ does. In strictly positive codimension the same axiom gives the following:

Proposition 7.7: The Twisted Cycle Class on Smooth Supports

Let $E_\bullet$ carry a twisted cycle-receiving structure and let α have gerbe class $[H]$. For each i, the subgroup of $H^{2i}(X;\mathbb{Z})$ generated by the classes of the $[Z,\mathcal{F}]_E$ is

$$\sum_Z \operatorname{ind}(\alpha|_Z)\cdot\mathbb{Z}\cdot\mathrm{cl}_i([Z]) \;\subseteq\; \ker d_3^{[H]}$$

the sum over smooth closed $Z \subseteq X$ of codimension i. For $i = 0$ it is $\operatorname{ind}(\alpha)\cdot\mathbb{Z}$.

Proof **:**
By axiom (4) the class of $[Z,\mathcal{F}]_E$ is $\mathrm{rk}(\mathcal{F})\cdot\mathrm{cl}_i([Z])$, and as $\mathcal{F}$ ranges over $\mathrm{Vect}_{[H]|_Z}(Z)$ its rank ranges over the subgroup of $\mathbb{Z}$ generated by the ranks of $\alpha|_Z$-twisted bundles, which is $\operatorname{ind}(\alpha|_Z)\cdot\mathbb{Z}$ by the definition of the index. Each such class is a permanent cycle, hence killed by $d_3^{[H]}$. For $i = 0$ the only Z is X itself and $\mathrm{cl}_0([X]) = 1$.

The invariant the structure detects in codimension i is thus the index of α on the supporting subvarieties rather than the algebraicity of untwisted classes: each subvariety contributes its classical cycle class weighted by $\operatorname{ind}(\alpha|_Z)$, which is the sense in which twisted K-theory records whether the Brauer class splits along a subvariety.

This describes the smooth-support part of the topological filtration on $K_0(X,\alpha)$, for the same reason that definition 3.1 is formulated on cycles supported on smooth subvarieties. The full categorical integral Hodge conjecture formulated by Perry [40] and investigated in the period-index context by Hotchkiss and Hotchkiss-Perry [24, 25] concerns the full algebraic group $K_0(X,\alpha) = K_0(\mathrm{Coh}(X,\alpha))$ and its image under the twisted Chern character. Because general α-twisted coherent sheaves may have singular supports or arise via non-split extensions that do not decompose into pushforwards of twisted bundles from smooth subvarieties, the cycle-receiving framework of this paper identifies the topological obstruction on smooth supports, leaving open the contribution of singular or non-split sheaf extensions.

Remark 7.8 (Rational twisted Chern character)*.* The class $[H]$ is torsion, so $[H]\otimes\mathbb{Q} = 0$, the rationalized twisted differential $d_3^{[H]}\otimes\mathbb{Q}$ vanishes as $\mathrm{Sq}^3_{\mathbb{Z}}\otimes\mathbb{Q}$ does, and the twisted Chern character is a rational isomorphism onto twisted rational cohomology [8, Prop. 7.4], which for a torsion class is identified non-canonically with ordinary even cohomology $H^{2*}(X;\mathbb{Q})$. The endomorphism algebra $\mathcal{E}nd(E)$ of an α-twisted sheaf is untwisted, so its characteristic classes are ordinary ones.

Two further points record how much a twist achieves when in the right framework. First, the twisted problem does produce failures of integrality. For X a very general product of a curve of genus $g \geq 2$ with elliptic curves and each prime ℓ, Hotchkiss exhibits a class $\alpha_\ell \in \mathrm{Br}(X)[\ell]$ for which the integral Hodge conjecture fails both for $D_{\mathrm{perf}}(X,\alpha_\ell)$ and for every Severi-Brauer variety of class α_ℓ, and for an abelian threefold with $\mathrm{per}(\alpha) = 2 < \operatorname{ind}(\alpha)$ it fails for every Severi-Brauer variety of class α [24, Thm. 1.5, Cor. 6.5, Thm. 1.6]. De Jong and Perry obtain integral Tate counterexamples by the same Hodge-theoretic route, conditionally on the Lefschetz standard conjecture [26]. Second, even where the twisted differential *is* the invariant, the sharpening a degree-three twist gives is bounded and internal to K-theory. The setting where such a twist demonstrably improves a bound is the topological period-index problem, an invariant of twisted K-theory itself: on spin^c six-manifolds Crowley and Grant prove $\operatorname{ind}(\alpha) \mid \mathrm{per}(\alpha)^2$, sharpening the general bound $\operatorname{ind} \mid \gcd(\mathrm{per},2)\,\mathrm{per}^2$ of Antieau-Williams [5] by a factor of two rather than reaching $\operatorname{ind} \mid \mathrm{per}$, and they show that the statement fails for general six-manifolds [14, Thms. 1.1, 1.2, 1.5]. These are statements within twisted K-theory, and by theorem 4.3 nothing analogous is available in cobordism, since the twist does not reach the differential there.

8 Conclusion

Twisting appears to add objects with which to detect algebraicity: the untwisted obstruction leaves classes undecided, the c_3 of section 5 among them, and twisted K-theory is built to see torsion in H^3. This paper shows it adds none. Twisted K-theory receives no cycle (proposition 3.7), and more generally any homotopy-commutative ring spectrum carrying a cycle-receiving structure is complex-oriented, and its structure is the Gysin system of that orientation (theorem 3.2), so no such theory refines cl_i beyond MU. Twisting the oriented theory MU changes nothing, since by Hertl's theorem every H^3-twist of MU is a ghost and the primary cobordism obstruction is undeformed (theorem 4.3).

Taking the twisted differential on its own as an operation on integral cohomology is no better: on the algebraic classes it either fails to vanish, in which case it is no obstruction (corollary 4.7), or it agrees there with the untwisted differential. Restricted to a single class the condition is a divisibility $N \big| d$ in which d is algebro-geometric data (proposition 4.10). On $B(\mathrm{SL}_6/\mu_2)$ the twisted obstruction on the degree-six Chern class is $o_{[H]}(\tilde{c}_3) \in \{0, [H]^3\}$ (corollary 5.5), and its admissibility on $\mathbb{Z}\tilde{c}_3$ is equivalent to the non-algebraicity of $\tilde{c}_3$ (proposition 5.8), which is the question the integral Hodge conjecture asks there. The same mechanism runs for the family SL_n/μ_2, n even (theorem 5.7).

What decides algebraicity is not this differential but the finer information of $\widetilde{\mathrm{cl}}$: the cokernel of the realization $\Omega^i(X) \to MU^{2i}(X(\mathbb{C})) \otimes_{MU^*} \mathbb{Z}$ of Levine-Morel algebraic cobordism into complex cobordism (proposition 6.1), a class being unobstructed exactly when it is realized by an algebraic cycle (corollary 6.2). That group is an invariant of the algebraic side, and by theorem 3.2 no complex-oriented refinement of the cycle map can see it.

The twisted differential is therefore no new obstruction to the untwisted conjecture. It is the governing obstruction for the twisted one (section 7), where the objects tested for algebraicity are themselves twisted sheaves, and the invariant that replaces the unit is the index of the Brauer class, tied to the period by the same degree-zero corner of the spectral sequence that made the unit fail here (proposition 7.6). That conjecture is logically independent of the untwisted one, and the corner separates them: $E_\infty^{0,0}$ always contains $\mathrm{ind}(\alpha)$ and contains 1 exactly when the twist vanishes. The unit that the untwisted conjecture requires is the one thing a nonzero twist cannot supply, and the index that replaces it is the invariant the twisted conjecture studies.

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